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BAXI



Installation, user and service manual

Reversible monobloc air-to-water heat pump

Outdoor unit

Assure HP55-4-1-PHMB
Assure HP55-6-1-PHMB
Assure HP55-8-1-PHMB
Assure HP55-11-1-PHMB
Assure HP55-11-3-PHMB
Assure HP55-13-1-PHMB
Assure HP55-13-3-PHMB

Please keep these instructions in a safe place.
If you move house, please hand them over to the next occupier.

Contents

1 Safety instructions 4

2 Standard delivery 6

3 Symbols used 7
3.1 Symbols used in the manual 7
3.2 Symbols used on the outdoor unit 7
3.3 Symbols used on the data plate 8

4 Technical specifications 8
4.1 Technical data 8
4.1.1 Outdoor unit technical specifications 8
4.1.2 Space heaters with medium-temperature heat pump 11
4.1.3 Available pressure 13
4.2 Sensor specification 15
4.3 Dimensions and connections 16
4.3.1 Outdoor units Assure HP55-4-1-PHMB - Assure HP55-6-1-PHMB 16
4.3.2 Assure HP55-8-1-PHMB outdoor unit 17
4.3.3 Outdoor units Assure HP55-11-1-PHMB - Assure HP55-11-3-PHMB - Assure HP55-13-1-PHMB - Assure HP55-13-3-PHMB 18
4.4 Refrigerant diagram 19
4.5 Electrical diagram 21

5 Description of the product 25
5.1 Data plate 25
5.2 Main components of the outdoor unit 25
5.3 Terminal blocks 27

6 Installation 29
6.1 Checking the condition of the packaging of the outdoor unit 29
6.2 Selecting the location of the outdoor unit 30
6.2.1 Protective perimeter 30
6.2.2 Minimum distances from the walls 31
6.2.3 General rules 33
6.2.4 Exposure to strong winds 33
6.2.5 Installation in coastal areas 34
6.2.6 Exposure to strong solar radiation 34
6.2.7 Installation in cold climates 34
6.2.8 Managing acoustic comfort 34
6.3 Preparing for installation 35
6.3.1 Condensate drainage 36
6.3.2 Preparing a ground installation 36
6.4 Unpack the outdoor unit 37
6.5 Transport 37
6.5.1 Transporting the outdoor unit using a crane 37
6.5.2 Transporting the outdoor unit without a crane 38
6.6 Putting the outdoor unit in place 39
6.6.1 Removing the exchanger protection 39
6.6.2 Remove the compressor support from the outdoor units Assure HP55-8-1-PHMB - Assure HP55-11-1-PHMB - Assure HP55-13-1-PHMB - Assure HP55-11-3-PHMB - Assure HP55-13-3-PHMB 39
6.6.3 Connecting the condensate drain hose 40
6.7 Hydraulic connections 40
6.7.1 Minimum volume of water 40
6.7.2 Heating water specifications 41
6.7.3 Volume of the expansion vessel 41
6.7.4 Flow rate range 42
6.7.5 Connecting the outdoor unit to the hydraulic circuit 42
6.7.6 Water pipe insulation 43
6.7.7 Drying the flow switch 43
6.8 Freeze protection 44
6.8.1 Software protection 44
6.8.2 Frost protection 44
6.9 Electrical connections 45
6.9.1 Checking and preparing the electrical installation 45

6.9.2 Precautions for electrical wiring . . . 46
6.9.3 Accessing the main control box . . . 46
6.9.4 Connecting the electrical circuits . . . 47
6.9.5 Cable routing . . . 47
6.9.6 Connecting the outdoor unit to the power supply . . . 49
6.9.7 Connect the outdoor unit to the indoor unit . . . 50
6.9.8 Connection for Other Optional Components . . . 50
6.10 **Refit the outdoor unit panels** . . . 51
6.11 Sealing building entry points . . . 51
6.12 Flushing and filling the heating installation with mains water . . . 51
6.12.1 Flushing new installations and installations less than 6 months old . . . 51
6.12.2 Flushing an existing installation . . . 51
6.12.3 Filling the heating circuit with mains water . . . 52
6.13 Flushing and filling a heating installation with glycol water . . . 52
6.13.1 Flushing an installation before filling with glycol water . . . 52
6.13.2 Filling the heating circuit with glycol water . . . 52
6.13.3 Filling the heating circuit with glycol water . . . 53
6.14 Checking water circuit . . . 54

7 User instructions . . . 54
7.1 Precautions regarding the location of the outdoor unit . . . 54
7.2 Prolonged power outage in winter . . . 54
7.3 Clearing the snow accumulated around the outdoor unit . . . 54
7.4 Cleaning the outdoor unit . . . 54

8 Maintenance . . . 55
8.1 Safety precautions for maintenance . . . 55
8.2 Maintenance checklist . . . 55
8.3 Service information . . . 56
8.3.1 Label for refrigerant presence . . . 56
8.3.2 Leak detection methods . . . 56
8.3.3 Check of refrigeration equipment . . . 56
8.3.4 Check of electrical devices . . . 57
8.3.5 Pre-pressure adjustment of the expansion vessel . . . 57

9 Error codes . . . 57

10 Decommissioning and disposal . . . 60
10.1 Disposal and recycling . . . 60
10.2 Brine disposal/recycling . . . 60
10.3 Recover refrigerants . . . 61
10.4 Labelling . . . 61
10.5 Recovery equipment . . . 61

11 Appendix . . . 62
11.1 Benchmark Commissioning & Warranty Validation Service Record . . . 62
11.2 Service record . . . 63
11.3 Product data sheet . . . 65
11.4 Servicing information . . . 66

1 Safety instructions

General safety instructions
This appliance can be used by children aged from 8 years and above and persons with reduced physical, sensory or mental capabilities or lack of experience and knowledge if they have been given supervision or instruction concerning use of the appliance in a safe way and understand the hazards involved. Children shall not play with the appliance. Children must not carry out any unsupervised cleaning or maintenance operations.

Before any work, read the documents provided with the appliance carefully. These documents are also available on our website. See the back cover.

Keep these documents close to the place where the appliance is installed.

Only qualified professionals are authorised to carry out installation, commissioning, maintenance, repair or removal work on the appliance. They must follow the prevailing local and national regulations.

Do not make any modifications to the appliance without the written consent of the manufacturer. To benefit from warranty cover, no modifications should be made to the appliance.

Installation location
The outdoor unit contains a category A3 flammable refrigerant. In the event of a leak, this refrigerant can form a flammable or explosive atmosphere. Specific rules must be respected within the defined protective zone in the immediate proximity of the outdoor unit.



See
Installation chapter, Protective perimeter.

Follow all the instructions in the Installation chapter.

In addition:

- Protect the outdoor unit from any accumulation of snow.
- Do not install at an altitude of more than 2000 metres above sea level
- Do not install in a location exposed to combustible gas

In coastal areas, the salt or corrosive air or sulphate gases in the environment may cause corrosion which can shorten the lifespan of the outdoor unit.

Refrigerant circuit
The outdoor unit contains a highly flammable refrigerant.

Respect the national regulations concerning refrigerants.

All work on the refrigerant circuit must be done by a qualified professional, according to prevailing codes of practice and safety in the profession (recovery of the refrigerant, brazing under nitrogen). All brazing work must be done by qualified brazers.

By qualified professional, we mean a person who has the qualifications pertaining to this refrigerant handling and piping work as stipulated by the local laws and regulations, and who has been trained in matters relating to refrigerant handling and piping work.

When installing, relocating, or servicing the heat pump, only use the specified refrigerant R290 to charge the refrigerant lines. Do not mix the R290 refrigerant with any other refrigerant and do not allow air, liquids or other gases to remain in the lines.

Repair the refrigerant leak before starting up the installation once more.



Caution
Do not attempt to speed up the defrosting process or clean using methods other than those recommended by the manufacturer.
The appliance must be stored in a location which does not contain any continuously operating ignition sources (for example: naked flames, operating gas appliance or electric radiator).
Do not pierce or burn.
Be aware that refrigerants may not contain an odour.

Heating circuit
Respect the minimum and maximum water pressure and temperature to ensure the heating system operates correctly. See the Technical Specifications chapter.

Instructions for using glycol water
Ensure that all the installation components are compatible with the use of glycol water.

Only use high-quality food-grade glycol water solutions based on monopropylene glycol (MPG).

Use a monopropylene glycol solution concentration which is compatible with the appliance. See the chapter on Precautions to take when filling with glycol water.

Electrical connections

Only a qualified installer or a qualified technician is authorised to work on the appliance's electrical system, as an incorrectly performed intervention may cause electrical shocks and/or electrical leakage.

Install the appliance in accordance with national rules on electrical installation.

In order to avoid the danger of an unexpected thermal circuit breaker reset, this appliance must not be powered through an external switch, such as a timer, or be connected to a circuit which is regularly switched on and off by the electricity provider.

The appliance is intended to be permanently connected to the mains. A disconnection device must be fitted to the fixed wiring in accordance with the installation rules.

Before any wiring work on the electrical circuit, switch off the power supply, check that no voltage is present and secure the circuit breaker with a circuit breaker lockout.

Use wiring that meets the specifications in the installation manual and local regulations and laws. Use of wiring which does not meet the specifications may give rise to electric shocks, electrical leakage, smoke and/or a fire.

Power supply cables outside buildings must be thicker than polychloroprene-sheathed flexible cables (60245 IEC 57 cable).

This appliance must be connected to the protective earthing in accordance with prevailing installation standards. Earth the appliance before making any electrical connections. Incomplete grounding can cause a malfunction or electric shock.

To avoid electric shock, check that the length of the conductors between the cable clamp and the terminal blocks is such that the active conductors are put under tension before the earth conductor.

Install a circuit breaker that meets the specifications in the installation manual and local regulations and laws.

If a power supply cable comes with the appliance and it turns out to be damaged, it must be replaced by the manufacturer, its after-sales service or persons with similar qualifications in order to avoid danger.

Separate the extra-low voltage cables from the 230/400 V power supply cables.

See the Electrical connections chapter for the following operations:

- Choosing the type and calibre of the protective equipment
- Connection to the electrical network
- Wiring the appliance

Maintenance and repair work

Remove the casing only to perform maintenance and repair work. Put the casing back in place after maintenance and repair work.

Before starting work on the refrigerant circuit, switch off the appliance and wait a few minutes. Certain items of equipment such as the compressor and the pipes can reach temperatures in excess of 100 °C and high pressures, which may cause serious injuries.

Prior to beginning work on systems containing flammable refrigerants, safety checks are necessary to ensure that the risk of ignition is minimised.

Work shall be undertaken under a controlled procedure so as to minimise the risk of a flammable gas or vapour being present while the work is being performed.

All maintenance staff and others working in the local area shall be instructed on the nature of work being carried out. Work in confined spaces shall be avoided.

The area shall be checked with an appropriate refrigerant detector prior to and during work, to ensure the technician is aware of potentially flammable or explosive atmospheres.

If a refrigerant leak is detected, all naked flames shall be removed/extinguished. If a leakage of refrigerant is found which requires brazing, all of the refrigerant shall be recovered from the system prior to brazing activities.

If any hot work is to be conducted on the refrigerating equipment or any associated parts, appropriate fire extinguishing equipment shall be kept close by. Have a dry powder or CO_2 fire extinguisher adjacent to the charging area.

Do not smoke during maintenance operations.

Repair and maintenance of electrical components must include initial safety checks and component inspection procedures. In the event of a fault that could compromise safety, no power supply may be connected to the circuit until said fault has been dealt with satisfactorily. If the fault cannot be corrected immediately but it is necessary to continue operation, an appropriate temporary solution must be adopted. This shall be reported to the owner of the equipment to ensure all parties are duly informed.

Initial safety checks shall include:

- Discharging of the capacitors: this shall be done in a safe manner to avoid any possibility of sparking
- Ensuring no live electrical components and wiring are exposed while charging, recovering or draining the system
- Ensuring the continuity of the earth bonding.

Before any work, switch off the power supply for all the components of your installation.

Use only original spare parts.

Guidelines for the user
If you do not need to heat your home for a long period, deactivate the heating mode. Do not switch off the heat pump in order to guarantee the frost protection of the installation.

If you do need to switch off the heat pump and if there is a risk that the temperature inside and/or outside the building will fall below zero:

- Installation filled with mains water: drain the water pipes of the installation (heat pump, underfloor heating, etc.) to prevent the system from freezing
- Installation filled with glycol water: the installation is protected against freezing

Keep the appliance accessible at all times to allow work to be performed.

Never remove or cover the labels and data plates affixed to appliances. These must be legible throughout the entire lifetime of the appliance.

Manufacturer's liabilities
Our products are manufactured in compliance with the requirements of the various applicable directives. They are therefore delivered with the CE marking and any documents necessary. In the interests of the quality of our products, we strive constantly to improve them. We therefore reserve the right to modify the specifications given in this document.

Our liability as manufacturer may not be invoked in the following cases:

- Failure to abide by the instructions on installing, commissioning and maintaining the appliance
- Failure to abide by the instructions on using the appliance
- Faulty or insufficient maintenance of the appliance

Responsibilities of the installer
The installer is responsible for the installation and initial commissioning of the appliance. The installer must observe the following instructions:

- Read and follow the instructions given in the manuals provided with the appliance
- Install the appliance in compliance with prevailing legislation and standards
- Carry out initial commissioning and any checks necessary
- Explain the installation to the user
- If maintenance is necessary, warn the user of the obligation to check the appliance and keep it in good working order
- Hand over all manuals to the user

Responsibilities of the user
To guarantee optimum operation of the system, you must abide by the following instructions:

- Read and follow the instructions given in the manuals provided with the appliance
- Call on a qualified professional to carry out installation and initial commissioning
- Get your installer to explain your installation to you
- Have the required inspections and maintenance carried out by a qualified installer
- Keep the instruction manuals in good condition close to the appliance

2 Standard delivery

Tab.1

Package	Contents
Outdoor unit	• An outdoor unit • An accessories bag containing: - Y filter - Condensate drain connector - Energy label - 3 cable clamps - 2 paper edge protectors • An installation, user and service manual
Magnetic filter	A magnetic filter

3 Symbols used

3.1 Symbols used in the manual

This manual uses various danger levels to draw attention to special instructions. We do this to improve user safety, to prevent problems and to guarantee correct operation of the appliance.

Danger
Risk of dangerous situations that may result in serious personal injury.

Danger of electric shock
Risk of electric shock.

Warning
Risk of dangerous situations that may result in minor personal injury.

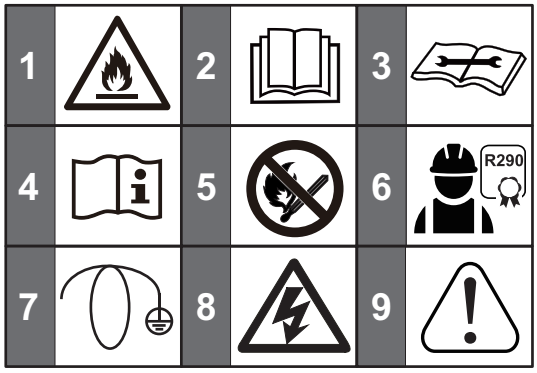
Caution
Risk of material damage.

Important
Please note: important information.

See
Reference to other manuals or pages in this manual.

3.2 Symbols used on the outdoor unit

Fig.1

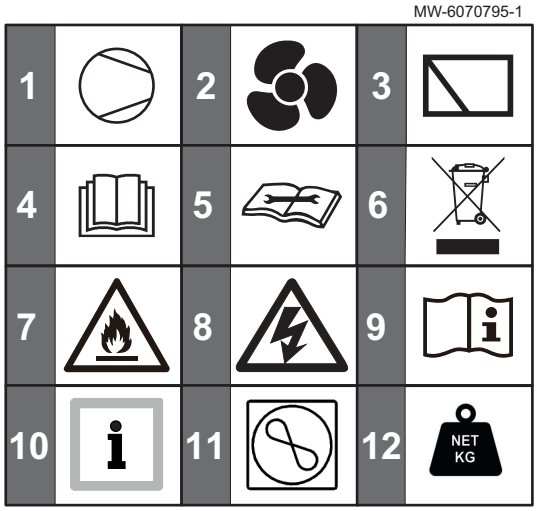


MW-6070683-2

1 The appliance contains a highly flammable refrigerant (A3)
2 Before installing and commissioning the appliance, carefully read the instruction manuals provided
3 Read the technical manual
4 See operating instructions
5 Naked flames prohibited
6 All work on the refrigeration circuit must be carried out by a qualified professional
7 To avoid electric shock, make sure that the length of the conductors between the traction arrester device and the terminal blocks is such that the active conductors are put under tension before the earth conductor
8 Caution: danger of electric shock
9 Important information

3.3 Symbols used on the data plate

Fig.2



1 Compressor
2 Fan
3 PCBs
4 Before installing and commissioning the appliance, carefully read the instruction manuals provided
5 Read the technical manual
6 Dispose of used products in an appropriate recovery and recycling structure
7 The appliance contains a highly flammable refrigerant (A3)
8 Caution: danger of electric shock
9 See operating instructions
10 Information
11 Information concerning the heat pump: refrigerant type, maximum allowable operating pressure and absorbed output
12 Net weight

4 Technical specifications

4.1 Technical data

4.1.1 Outdoor unit technical specifications

Tab.2 Outdoor unit conditions of use

Limit operating temperatures		Outdoor unit
Heating mode	Water	Maximum 75 °C
	Outdoor air	-25 °C to 35 °C
Cooling mode	Water	Maximum 25 °C
	Outdoor air	-5 °C to 46 °C
Domestic hot water by heat pump	Outdoor air	-25 °C to 46 °C

Tab.3 Outdoor unit specifications

Specifications	Unit	Assure HP55-4-1-PHMB	Assure HP55-6-1-PHMB	Assure HP55-8-1-PHMB	Assure HP55-11-1-PHMB	Assure HP55-11-3-PHMB	Assure HP55-13-1-PHMB	Assure HP55-13-3-PHMB
Minimum flow rate required	l/min	6.67	6.67	6.67	11.67	11.67	11.67	11.67
Erp sound power level	dB	48	48	49	52	52	52	52
Weight								
Net weight	kg	94	94	121.5	139.5	141.5	139.5	141.5
Gross weight	kg	114	114	143.5	161.5	163.5	161.5	163.5
Connections								
Water inlet/outlet	-	G1"BSP	G1"BSP	G1 1/4"BSP	G1 1/4"BSP	G1 1/4"BSP	G1 1/4"BSP	G1 1/4"BSP
Water drain	-	Hose connection	Hose connection	Hose connection	Hose connection	Hose connection	Hose connection	Hose connection
Expansion vessel								
Volume	l	8	8	8	8	8	8	8
Maximum operating pressure	MPa (bar)	0.8 (8)	0.8 (8)	0.8 (8)	0.8 (8)	0.8 (8)	0.8 (8)	0.8 (8)
Pump								
Type		Water cooled	Water cooled	Water cooled	Water cooled	Water cooled	Water cooled	Water cooled

Specifications	Unit	Assure HP55-4-1-PHMB	Assure HP55-6-1-PHMB	Assure HP55-8-1-PHMB	Assure HP55-11-1-PHMB	Assure HP55-11-3-PHMB	Assure HP55-13-1-PHMB	Assure HP55-13-3-PHMB
Number of speeds		Variable speed	Variable speed	Variable speed	Variable speed	Variable speed	Variable speed	Variable speed
Safety pressure relief valve in water circuit	MPa (bar)	0.3 (3)	0.3 (3)	0.3 (3)	0.3 (3)	0.3 (3)	0.3 (3)	0.3 (3)
Type of refrigerant	-	R290	R290	R290	R290	R290	R290	R290
Refrigerant charge	kg	0.7	0.7	1.1	1.25	1.25	1.25	1.25
Refrigerant charge(1)	tCO_2e	0.0021	0.0021	0.0033	0.00375	0.00375	0.00375	0.00375

(1) The quantity of refrigerant in CO_2 equivalent is calculated using the following formula: quantity (in kg) of refrigerant x GWP / 1000. The Global Warming Potential (GWP) of R290 is 3.

Tab.4 Heating mode: outdoor air temperature +2 °C, outlet water temperature +35 °C.

Measurement type	Unit	Assure HP55-4-1-PHMB	Assure HP55-6-1-PHMB	Assure HP55-8-1-PHMB	Assure HP55-11-1-PHMB	Assure HP55-11-3-PHMB	Assure HP55-13-1-PHMB	Assure HP55-13-3-PHMB
Heating capacity at nominal point	kW	4.30	5.88	8.20	11.00	11.00	12.80	12.80
Heating capacity min/max	kW	2.53/6.50	2.53/7.39	3.84/10.61	5.07/13.68	5.07/13.68	5.07/16.60	5.07/16.60
Coefficient of performance (COP)	-	4.11	3.82	3.60	3.45	3.45	3.28	3.28
Absorbed electrical power	kWe	1.05	1.54	2.28	3.19	3.19	3.90	3.90

Tab.5 Heating mode: outdoor air temperature +2 °C, outlet water temperature +55 °C.

Measurement type	Unit	Assure HP55-4-1-PHMB	Assure HP55-6-1-PHMB	Assure HP55-8-1-PHMB	Assure HP55-11-1-PHMB	Assure HP55-11-3-PHMB	Assure HP55-13-1-PHMB	Assure HP55-13-3-PHMB
Heating capacity at nominal point	kW	4.40	5.75	7.75	11.20	11.20	13.00	13.00
Heating capacity min/max	kW	3.67/5.88	3.67/6.70	5.33/9.88	7.64/12.61	7.64/12.61	7.64/15.39	7.64/15.39
Coefficient of performance (COP)	-	2.72	2.66	2.56	2.57	2.57	2.46	2.46
Absorbed electrical power	kWe	1.62	2.16	3.03	4.36	4.36	5.29	5.29

Tab.6 Heating mode: outdoor air temperature +7 °C, outlet water temperature +35 °C. Performance certified as per EN 14511-2.

Measurement type	Unit	Assure HP55-4-1-PHMB	Assure HP55-6-1-PHMB	Assure HP55-8-1-PHMB	Assure HP55-11-1-PHMB	Assure HP55-11-3-PHMB	Assure HP55-13-1-PHMB	Assure HP55-13-3-PHMB
Heating capacity at nominal point	kW	4.40	6.26	8.50	11.50	11.50	13.50	13.50
Heating capacity min/max	kW	2.93/6.86	2.93/7.70	5.58/11.05	5.87/14.72	5.87/14.72	5.87/17.57	5.87/17.57
Coefficient of performance (COP)	-	5.17	4.89	4.98	4.85	4.85	4.60	4.60
Absorbed electrical power	kWe	0.85	1.28	1.71	2.37	2.37	2.94	2.94
Nominal water flow rate (ΔT = 5 K)	l/min	12.6	17.9	24.4	33	33	38.7	38.7

Tab.7 Heating mode: outdoor air temperature +7 °C, outlet water temperature +55 °C. Performance certified in accordance with EN 14511-2.

Measurement type	Unit	Assure HP55-4-1-PHMB	Assure HP55-6-1-PHMB	Assure HP55-8-1-PHMB	Assure HP55-11-1-PHMB	Assure HP55-11-3-PHMB	Assure HP55-13-1-PHMB	Assure HP55-13-3-PHMB
Heating capacity at nominal point	kW	4.40	6.10	8.00	11.50	11.50	13.50	13.50
Heating capacity min/max	kW	4.55/6.20	4.55/6.97	6.52/10.17	9.75/13.61	9.75/13.61	9.75/16.40	9.75/16.40
Coefficient of performance (COP)	-	3.24	3.12	3.18	3.15	3.15	3.04	3.04
Absorbed electrical power	kW	1.36	1.96	2.52	3.65	3.65	4.44	4.44
Nominal water flow rate (ΔT = 5 K)	l/min	12.6	17.5	22.9	33	33	38.7	38.7

Tab.8 Cooling mode: outdoor air temperature +35 °C, outlet water temperature +18 °C. Stated performances in accordance with EN 14511-2.

Measurement type	Unit	Assure HP55-4-1-PHMB	Assure HP55-6-1-PHMB	Assure HP55-8-1-PHMB	Assure HP55-11-1-PHMB	Assure HP55-11-3-PHMB	Assure HP55-13-1-PHMB	Assure HP55-13-3-PHMB
Cooling capacity at nominal point	kW	4.56	6.40	9.00	11.90	11.90	13.90	13.90
Min/max cooling capacity	kW	2.29/5.66	2.29/7.14	3.25/9.23	4.17/11.96	4.17/11.96	4.17/14.26	4.17/14.26
Energy efficiency ratio (EER)	-	5.49	5.12	4.95	4.52	4.52	4.22	4.22
Absorbed electrical power	kW	0.83	1.25	1.82	2.63	2.63	3.29	3.29
Nominal water flow rate (ΔT = 5 K)	l/min	12.6	17.9	24.4	33	33	38.7	38.7

Tab.9 Cooling mode: outdoor air temperature +35 °C, outlet water temperature +7 °C. Stated performances in accordance with EN 14511-2.

Measurement type	Unit	Assure HP55-4-1-PHMB	Assure HP55-6-1-PHMB	Assure HP55-8-1-PHMB	Assure HP55-11-1-PHMB	Assure HP55-11-3-PHMB	Assure HP55-13-1-PHMB	Assure HP55-13-3-PHMB
Cooling capacity at nominal point	kW	4.40	6.26	8.50	11.50	11.50	13.50	13.50
Min/max cooling capacity	kW	3.18/7.84	3.18/9.75	4.69/12.13	6.08/16.40	6.08/16.40	6.08/18.64	6.08/18.64
Energy efficiency ratio (EER)	-	3.69	3.20	3.28	3.05	3.05	2.80	2.80
Absorbed electrical power	kW	1.19	1.96	2.59	3.77	3.77	4.81	4.81
Nominal water flow rate (ΔT = 5 K)	l/min	13.1	18.3	25.8	34.1	34.1	39.8	39.8

Tab.10 Fuse – on printed circuit boards

	Main Control PCB	Inverter module PCB
Fuse name	FUSE-T-10A/250VAC-T-P	FUSE-T-30A/250VAC-T-P-HT
Working voltage (V)	250	250
Working current (A)	10	30

Tab.11 Flow switch

	Assure HP55-4-1-PHMB Assure HP55-6-1-PHMB Assure HP55-8-1-PHMB	Assure HP55-11-1-PHMB Assure HP55-11-3-PHMB Assure HP55-13-1-PHMB Assure HP55-13-3-PHMB
Setpoint	6 l/min ± 1	10 l/min ± 1

4.1.2 Space heaters with medium-temperature heat pump

Tab.12 Technical parameters for heat pump combination heaters (parameters declared for medium-temperature application: 55 °C)

Product name		Unit	Assure HP55-4-1-PHMB	Assure HP55-6-1-PHMB	Assure HP55-8-1-PHMB
Air-to-water heat pump	-	-	Yes	Yes	Yes
Water-to-water heat pump	-	-	No	No	No
Brine-to-water heat pump	-	-	No	No	No
Low-temperature heat pump	-	-	No	No	No
Equipped with a supplementary heater	-	-	No	No	No
Heat pump combination heater	-	-	No	No	No
Rated heat output under average conditions(1)	*Prated*	kW	4.90	6.10	7.80
Rated heat output under colder conditions	*Prated*	kW	4.30	5.90	8.00
Rated heat output under warmer conditions	*Prated*	kW	4.70	6.00	8.80
Declared capacity for heating for part load at an indoor temperature of 20 °C and outdoor temperature T_j					
T_j = -7 °C	*Pdh*	kW	4.42	5.40	6.95
T_j = +2 °C	*Pdh*	kW	2.72	3.13	4.09
T_j = +7 °C	*Pdh*	kW	2.55	2.58	3.47
T_j = +12 °C	*Pdh*	kW	3.01	3.02	4.03
T_j = bivalent temperature	*Pdh*	kW	4.42	5.40	6.95
T_j = operation limit temperature	*Pdh*	kW	4.80	5.37	7.71
Bivalent temperature	T_{biv}	°C	-7	-7	-7
Degradation coefficient(2)	*Cdh*	-	0.9	0.9	0.9
Seasonal space heating energy efficiency under average conditions	η_s	%	157	153	153
Seasonal space heating energy efficiency under colder conditions	η_s	%	124.3	132	136
Seasonal space heating energy efficiency under warmer conditions	η_s	%	170	179	188
Declared coefficient of performance or primary energy ratio for part load at an indoor temperature of 20 °C and outdoor temperature T_j					
T_j = -7 °C	*COPd*	-	2.59	2.40	2.36
T_j = +2 °C	*COPd*	-	3.94	3.79	3.83
T_j = +7 °C	*COPd*	-	4.94	5.15	5.05
T_j = +12 °C	*COPd*	-	6.44	6.53	6.41
T_j = bivalent temperature	*COPd*	-	2.59	2.40	2.36
T_j = operation limit temperature	*COPd*	-	2.27	2.25	2.02
Operation limit temperature for air-to-water heat pumps	*TOL*	°C	-10	-10	-10
Heating water operating limit temperature	*WTOL*	°C	75	75	75
Electrical power consumption					
Off mode	P_{OFF}	kW	0.009	0.009	0.009

Product name		Unit	Assure HP55-4-1-PHMB	Assure HP55-6-1-PHMB	Assure HP55-8-1-PHMB
Thermostat-off mode	P_{TO}	kW	0.010	0.010	0.009
Stand-by	P_{SB}	kW	0.009	0.009	0.009
Crankcase heater mode	P_{CK}	kW	0.000	0.000	0.000
Supplementary heater					
Rated heat output	*Psup*	kW	0.10	0.73	0.09
Type of energy input	-	-	Electricity	Electricity	Electricity
Other specifications					
Capacity control	-	-	Variable	Variable	Variable
Sound power level, indoors - outdoors	L_{WA}	dB	0 - 48	0 - 48	0 - 49
Annual energy consumption under average conditions	Q_{HE}	kWh	2535	3233	4140
Annual energy consumption under colder conditions	Q_{HE}	kWh	3328	4325	5659
Annual energy consumption under warmer conditions	Q_{HE}	kWh	1446	1762	2456
Rated air flow rate, outdoors for air-to-water heat pumps	-	m^3/h	2875	2875	4031

(1) The rated heat output *Prated* is equal to the design load for heating *Pdesignh*, and the rated heat output of a supplementary heater *Psup* is equal to the supplementary capacity for heating *sup(Tj)*.
(2) If *Cdh* is not determined by measurement, the default degradation coefficient is *Cdh* = 0.9.

Tab.13 Technical parameters for heat pump combination heaters (parameters declared for medium-temperature application: 55 °C)

Product name		Unit	Assure HP55-11-1-PHMB	Assure HP55-11-3-PHMB	Assure HP55-13-1-PHMB	Assure HP55-13-3-PHMB
Air-to-water heat pump	-	-	Yes	Yes	Yes	Yes
Water-to-water heat pump	-	-	No	No	No	No
Brine-to-water heat pump	-	-	No	No	No	No
Low-temperature heat pump	-	-	No	No	No	No
Equipped with a supplementary heater	-	-	No	No	No	No
Heat pump combination heater	-	-	No	No	No	No
Rated heat output under average conditions(1)	*Prated*	kW	12.10	12.10	13.70	13.70
Rated heat output under colder conditions	*Prated*	kW	10.80	10.80	13.90	13.90
Rated heat output under warmer conditions	*Prated*	kW	12.40	12.40	14.90	14.90
Declared capacity for heating for part load at an indoor temperature of 20 °C and outdoor temperature T_j						
T_j = -7 °C	*Pdh*	kW	10.88	10.88	11.87	11.87
T_j = +2 °C	*Pdh*	kW	6.56	6.56	7.37	7.37
T_j = +7 °C	*Pdh*	kW	4.78	4.78	4.87	4.87
T_j = +12 °C	*Pdh*	kW	5.83	5.83	5.83	5.83
T_j = bivalent temperature	*Pdh*	kW	10.88	10.88	11.87	11.87
T_j = operation limit temperature	*Pdh*	kW	10.71	10.71	11.20	11.20
Bivalent temperature	T_{biv}	°C	-7	-7	-7	-7
Degradation coefficient(2)	*Cdh*	-	0.9	0.9	0.9	0.9
Seasonal space heating energy efficiency under average conditions	η_s	%	147	147	146	146
Seasonal space heating energy efficiency under colder conditions	η_s	%	127	127	128	128
Seasonal space heating energy efficiency under warmer conditions	η_s	%	174	174	181	181

Product name		Unit	Assure HP55-11-1-PHMB	Assure HP55-11-3-PHMB	Assure HP55-13-1-PHMB	Assure HP55-13-3-PHMB
Declared coefficient of performance or primary energy ratio for part load at an indoor temperature of 20 °C and outdoor temperature T_j						
T_j = -7 °C	*COPd*	-	2.27	2.27	2.22	2.22
T_j = +2 °C	*COPd*	-	3.63	3.63	3.56	3.56
T_j = +7 °C	*COPd*	-	4.99	4.99	5.21	5.21
T_j = +12 °C	*COPd*	-	6.55	6.55	6.55	6.55
T_j = bivalent temperature	*COPd*	-	2.27	2.27	2.22	2.22
T_j = operation limit temperature	*COPd*	-	2.15	2.15	2.07	2.07
Operation limit temperature for air-to-water heat pumps	*TOL*	°C	-10	-10	-10	
Heating water operating limit temperature	*WTOL*	°C	75	75	75	75
Electrical power consumption						
Off mode	P_{OFF}	kW	0.010	0.010	0.010	0.010
Thermostat-off mode	P_{TO}	kW	0.015	0.015	0.015	0.015
Stand-by	P_{SB}	kW	0.010	0.010	0.010	0.010
Crankcase heater mode	P_{CK}	kW	0.000	0.000	0.000	0.000
Supplementary heater						
Rated heat output	*Psup*	kW	1.39	1.39	2.50	2.50
Type of energy input	-	-	Electricity	Electricity	Electricity	Electricity
Other specifications						
Capacity control	-	-	Variable	Variable	Variable	Variable
Sound power level, indoors - outdoors	L_{WA}	dB	0 - 52	0 - 52	0 - 52	0 - 52
Annual energy consumption under average conditions	Q_{HE}	kWh	6662	6662	7588	7588
Annual energy consumption under colder conditions	Q_{HE}	kWh	8197	8197	10408	10408
Annual energy consumption under warmer conditions	Q_{HE}	kWh	3724	3724	4306	4306
Rated air flow rate, outdoors for air-to-water heat pumps	-	m^3/h	4457	4457	5042	5042

(1) The rated heat output *Prated* is equal to the design load for heating *Pdesignh*, and the rated heat output of a supplementary heater *Psup* is equal to the supplementary capacity for heating *sup(Tj)*.
(2) If *Cdh* is not determined by measurement, the default degradation coefficient is *Cdh* = 0.9.

See
Back cover for contact details.

4.1.3 Available pressure

Important
The benchmark for the most efficient circulating pumps is EEI ≤ 0.20.

The relationship between the external static pressure and the water flow rate is shown as follows:

Fig.3 Assure HP55-4-1-PHMB - Assure HP55-6-1-PHMB

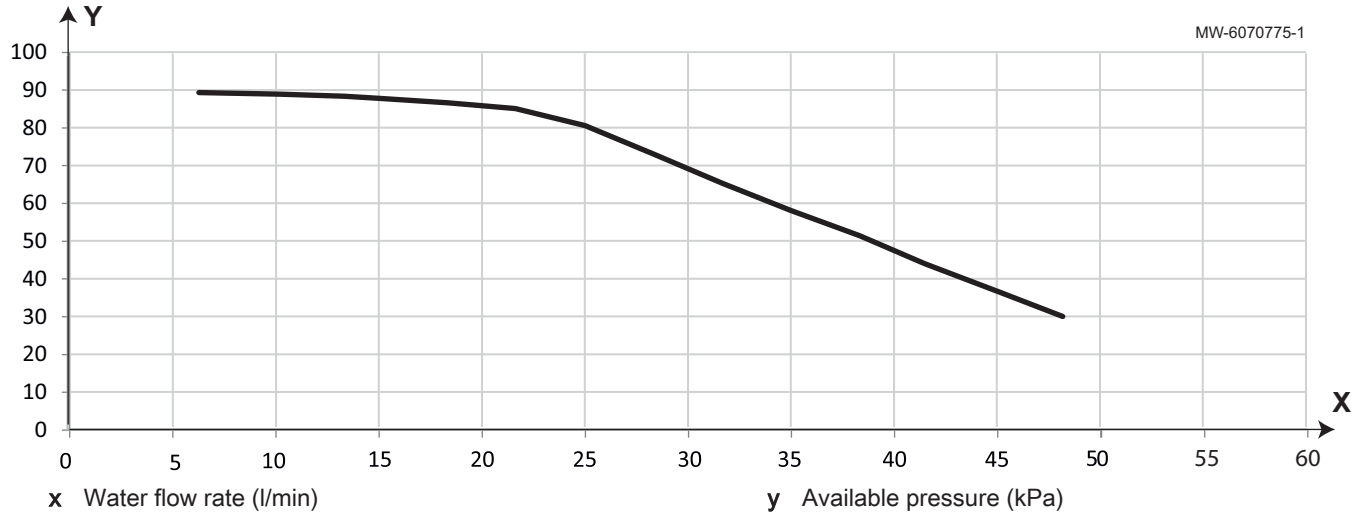


Fig.4 Assure HP55-8-1-PHMB

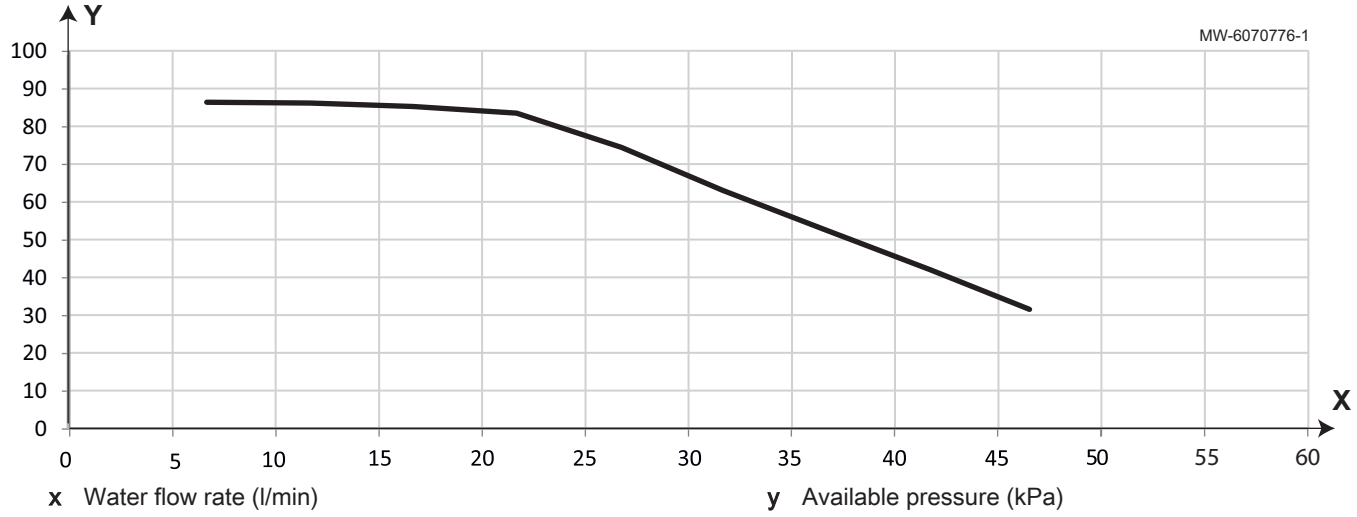
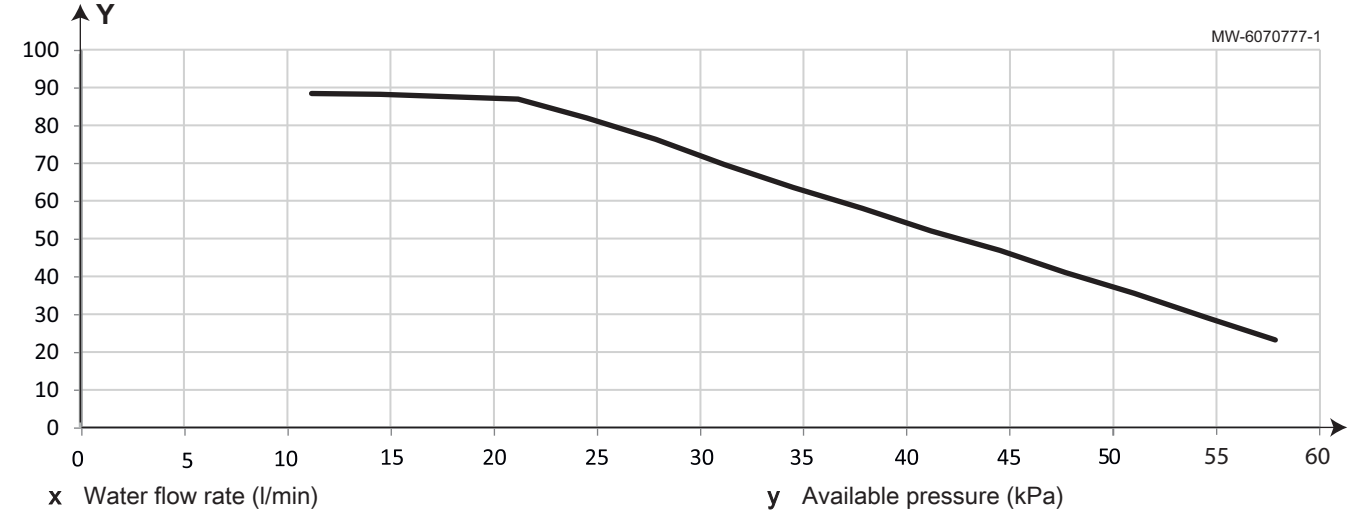


Fig.5 Assure HP55-11-1-PHMB - Assure HP55-11-3-PHMB - Assure HP55-13-1-PHMB - Assure HP55-13-3-PHMB



4.2 Sensor specification

Tab.14 Resistance characteristics of sensors in the water circuit

Temperature	°C	-10	0	10	20	30	40	50	60	70	80	90	100
Resistance [1]	kΩ	269.6	159.7	97.5	61.2	39.4	26	17.6	12.15	8.55	6.12	4.45	3.29

(1) The resistance tolerance is 3% at 50 °C and 5% at 25 °C.

The sensors in the water circuit are:

- Tw_in : heating return temperature sensor
- Tw_out: heating flow temperature sensor
- T1: water outlet temperature sensor downstream of the supplementary heater
- T5: domestic hot water cylinder temperature sensor
- Tw2: zone 2 water temperature sensor

Tab.15 Resistance characteristics of the compressor discharge temperature sensorTp

Temperature	°C	-20	-10	0	10	20	30	40	50
Resistance(1)	kΩ	516.71	301.63	180.41	110.64	69.60	44.89	29.66	20.03

Temperature	°C	60	70	80	90	100	110	120	130
Resistance(1)	kΩ	13.80	9.68	6.91	5	3.67	2.74	2.07	1.59

(1) The resistance tolerance is 4% at 100 °C and 5% at 75 °C.

Tab.16 Resistance characteristics of sensors in the refrigerant circuit

Temperature	°C	-20	-10	0	10	20	25	30
Resistance(1)	kΩ	106.73	59.78	34.60	20.61	12.64	10	7.97

Temperature	°C	30	40	50	60	70	80	90	100
Resistance(1)	kΩ	7.97	5.18	3.45	2.35	1.63	1.15	0.83	0.61

(1) The resistance tolerance is 5% at 50 °C and 3% at 25 °C.

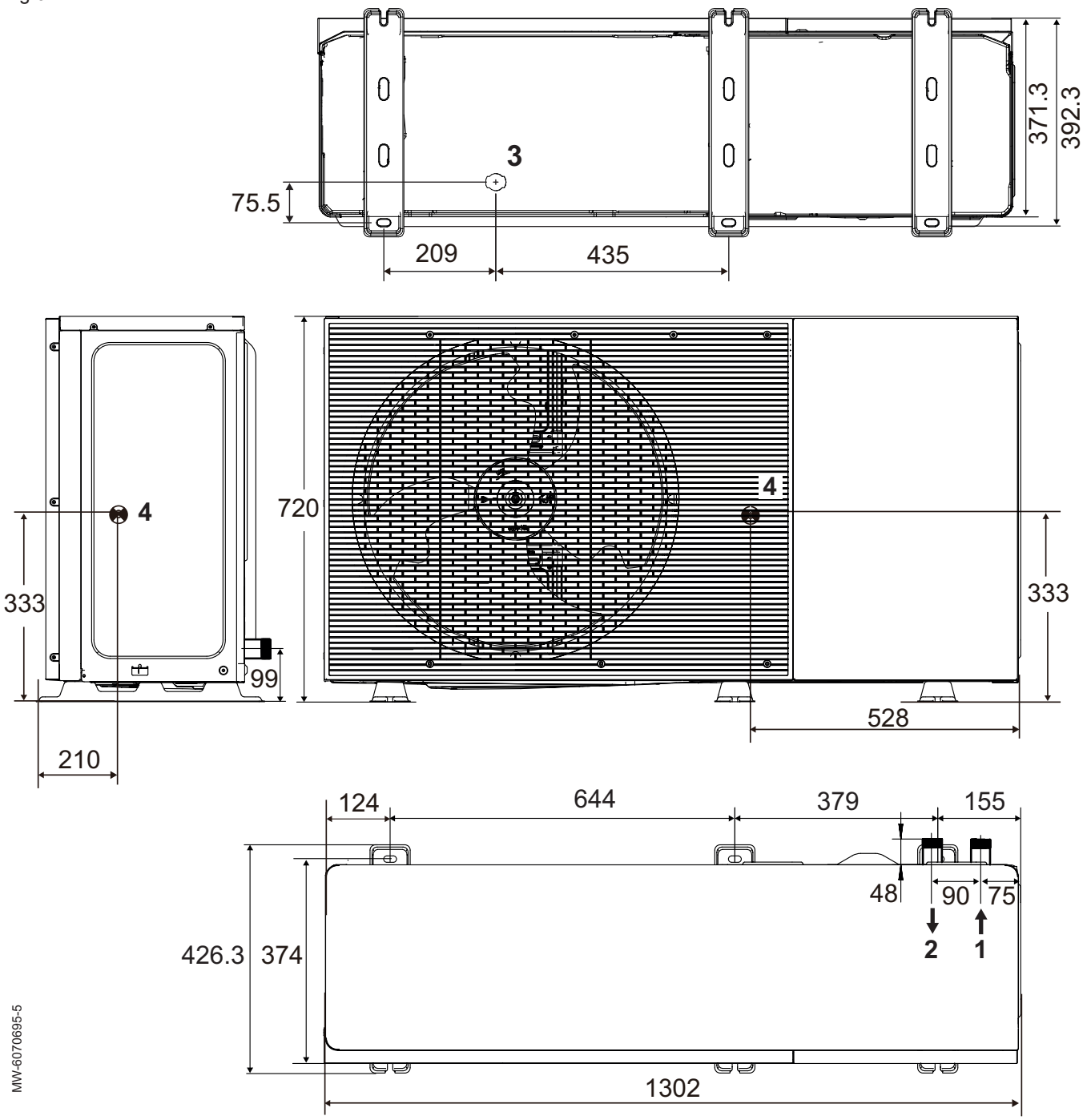
The sensors in the refrigerant circuit are:

- Th : compressor suction temperature sensor
- T4: outdoor air temperature sensor
- T3: heat exchanger temperature sensor
- TL: heat exchanger outlet refrigerant temperature sensor (for cooling mode)
- T2: refrigerant temperature sensor at the plate heat exchanger inlet (for cooling mode)
- T2B: plate heat exchanger outlet refrigerant temperature sensor (for cooling mode)

4.3 Dimensions and connections

4.3.1 Outdoor units Assure HP55-4-1-PHMB - Assure HP55-6-1-PHMB

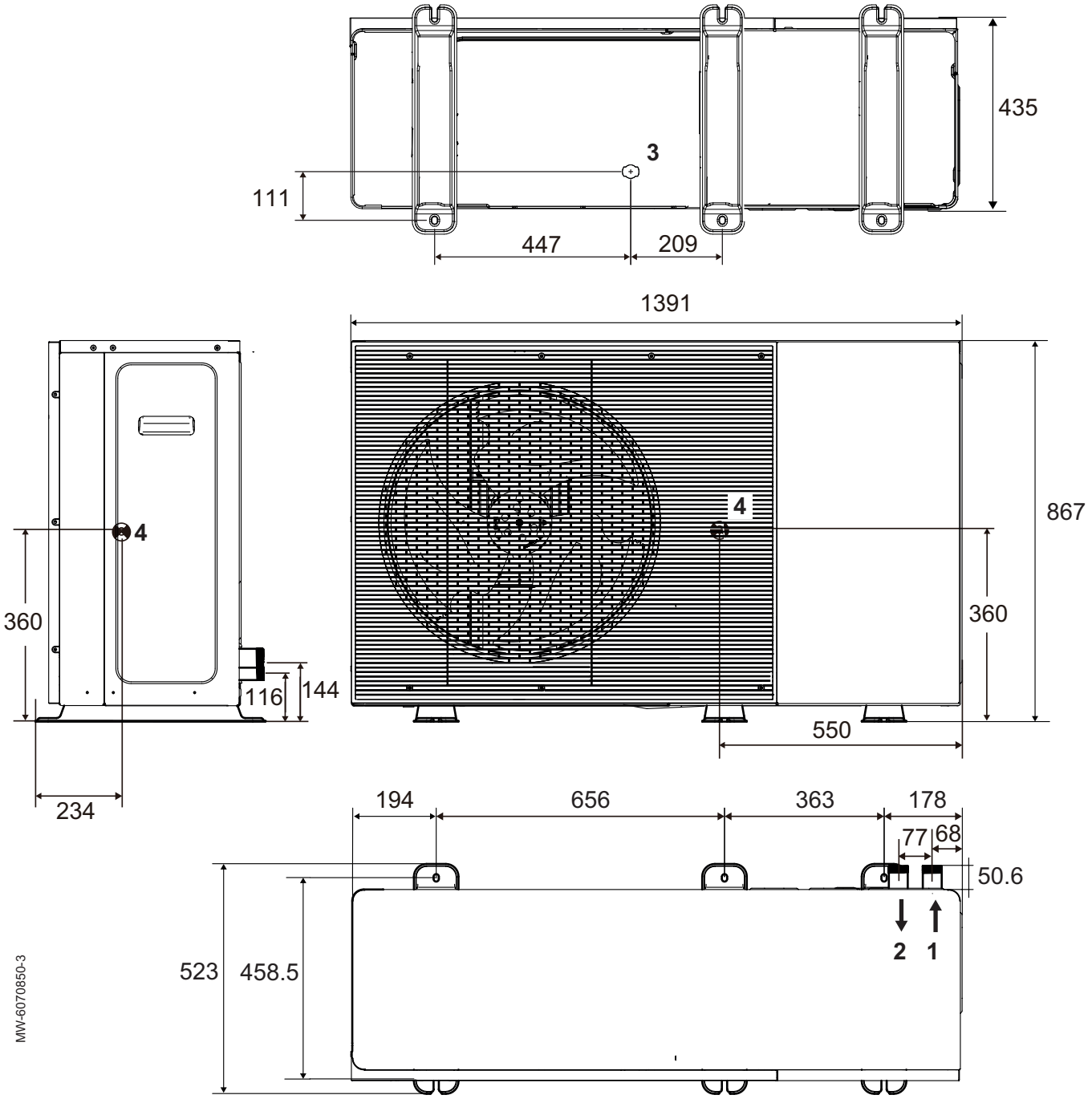
Fig.6



1 Heating circuit flow G1"BSP
2 Heating circuit return G1"BSP
3 Condensate drain hole
4 Centre of gravity

4.3.2 Assure HP55-8-1-PHMB outdoor unit

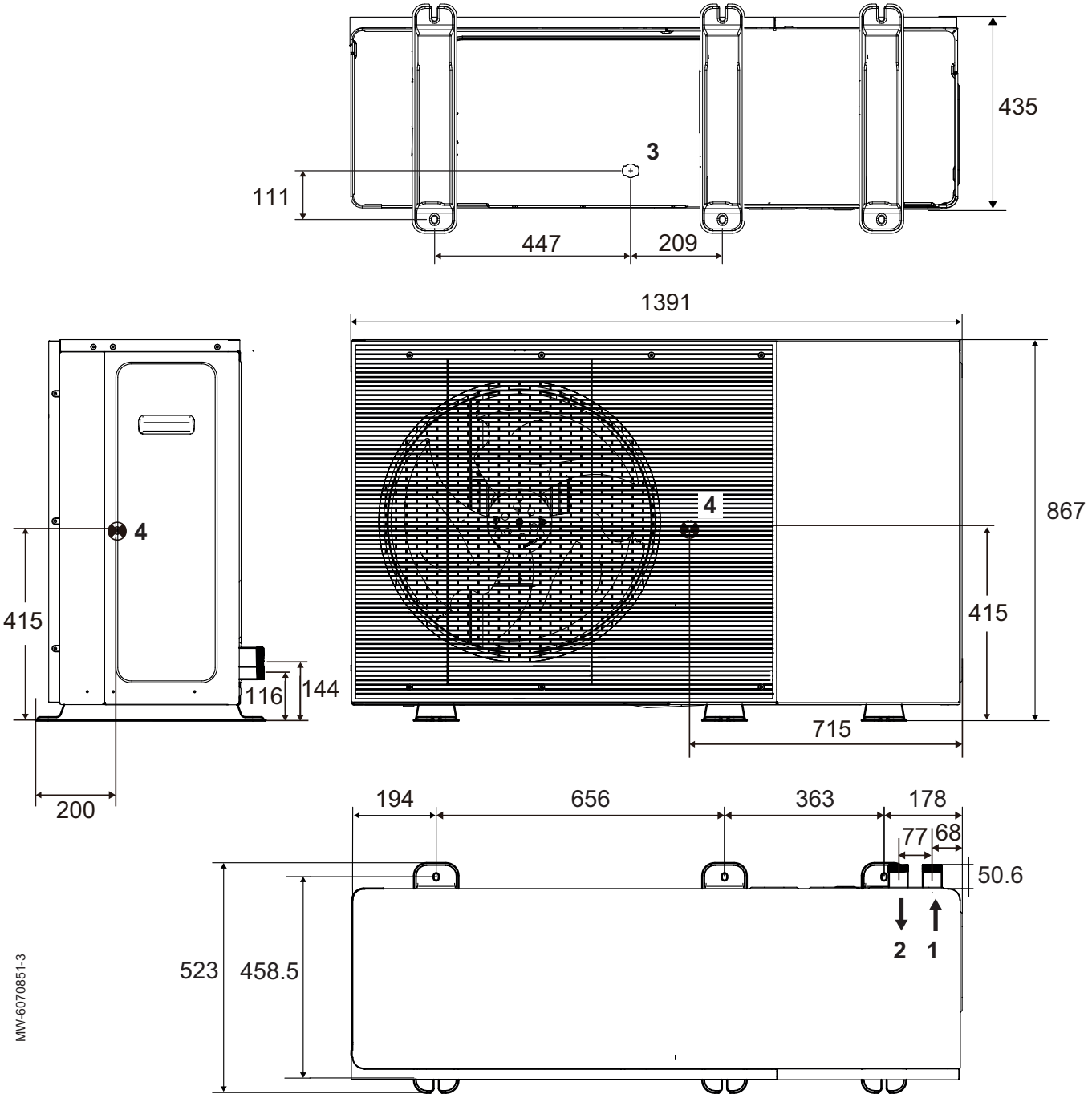
Fig.7



1 Heating circuit flow G1" 1/4"BSP
2 Heating circuit return G1"1/4" BSP
3 Condensate drain hole
4 Centre of gravity

4.3.3 Outdoor units Assure HP55-11-1-PHMB - Assure HP55-11-3-PHMB - Assure HP55-13-1-PHMB - Assure HP55-13-3-PHMB

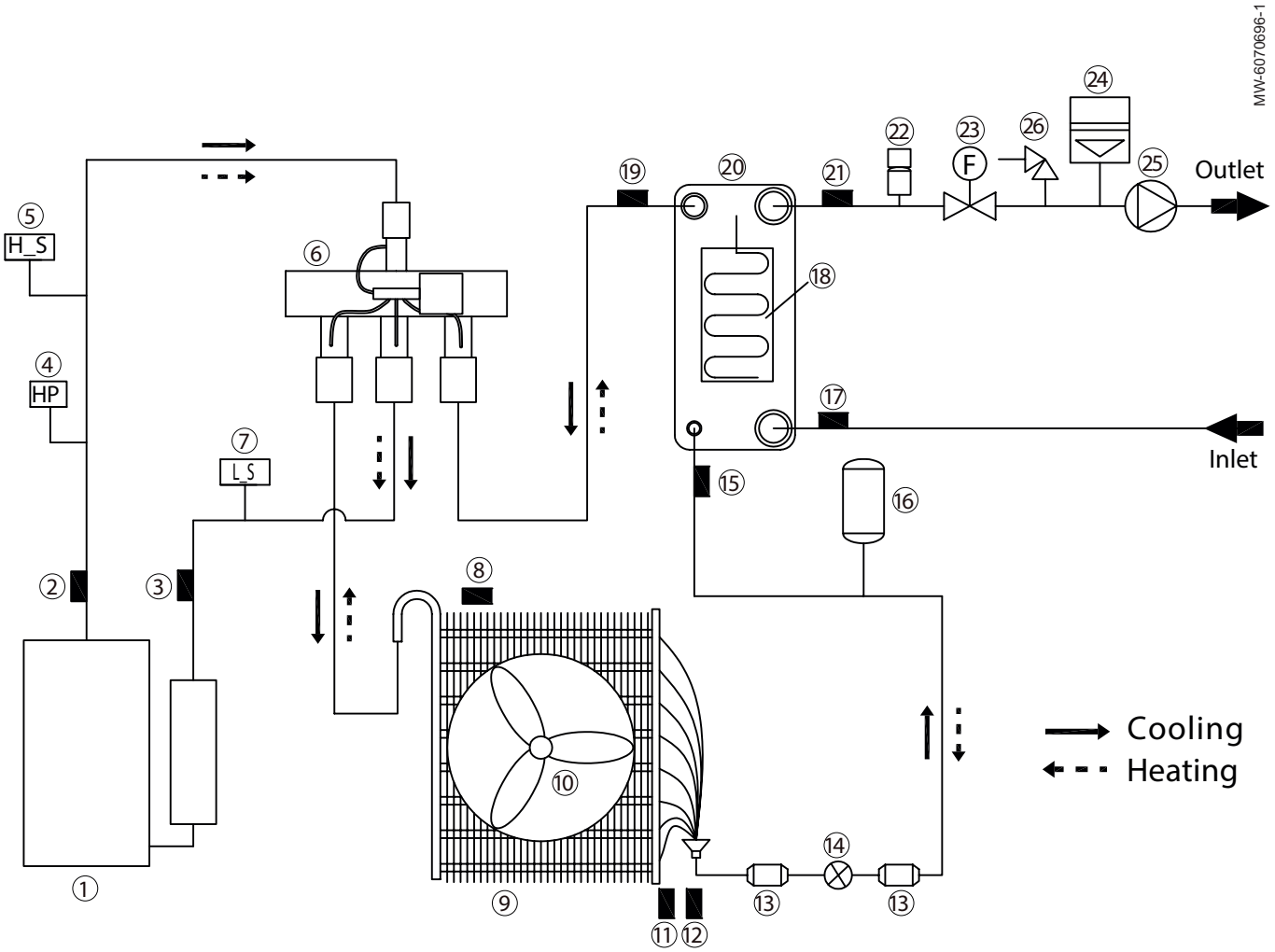
Fig.8



1 Heating circuit flow G1" 1/4"BSP
2 Heating circuit return G1"1/4" BSP
3 Condensate drain hole
4 Centre of gravity

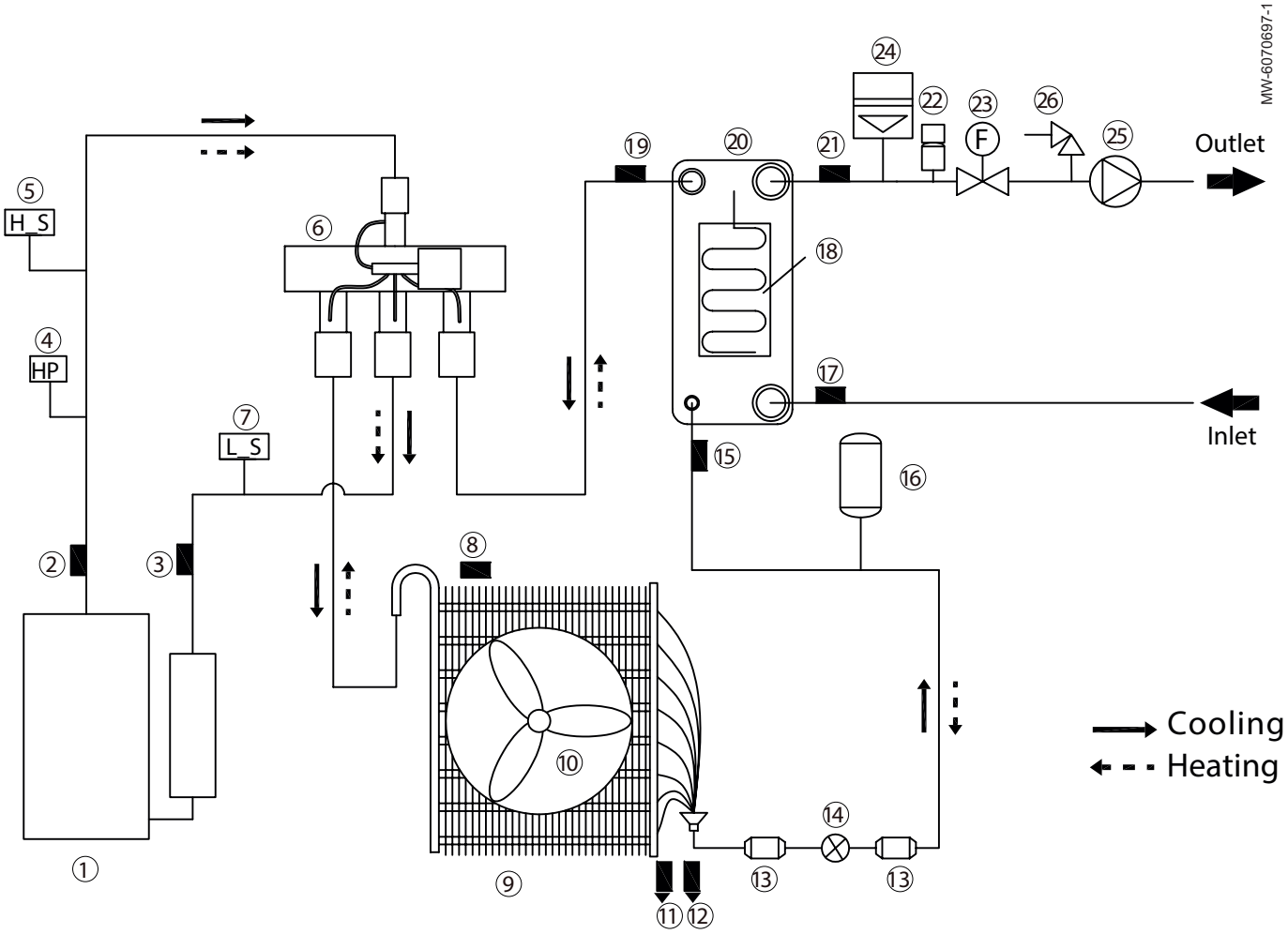
4.4 Refrigerant diagram

Fig.9 Assure HP55-4-1-PHMB - Assure HP55-6-1-PHMB



1 Compressor
2 Compressor discharge temperature sensor (high-pressure) Tp
3 Compressor intake temperature sensor (low pressure) Th
4 High-pressure switch
5 High-pressure sensor
6 4-way valve
7 Low pressure sensor
8 Outdoor air temperature sensor T4
9 Finned heat exchanger (evaporator in heating mode)
10 Fan
11 Heat exchanger temperature sensor T3
12 Heat exchanger outlet refrigerant temperature sensor (in cooling mode) TL
13 Filter
14 Electronic expansion valve
15 Refrigerant temperature sensor at the plate heat exchanger (in cooling mode) T2
16 Liquid receiver
17 Heating return temperature sensor Tw_in
18 Heat tape (plate heat exchanger)
19 Plate heat exchanger outlet refrigerant temperature sensor (in cooling mode) T2B
20 Plate heat exchanger (condenser in heating mode)
21 Heating flow temperature sensor Tw_out
22 Automatic bleed valve
23 Flow switch
24 Expansion vessel
25 Circulating pump
26 Safety pressure relief valve
Outlet Outlet - Heating circuit water flow
Inlet Inlet - Heating circuit water return
Cooling Cooling mode
Heating Heating mode

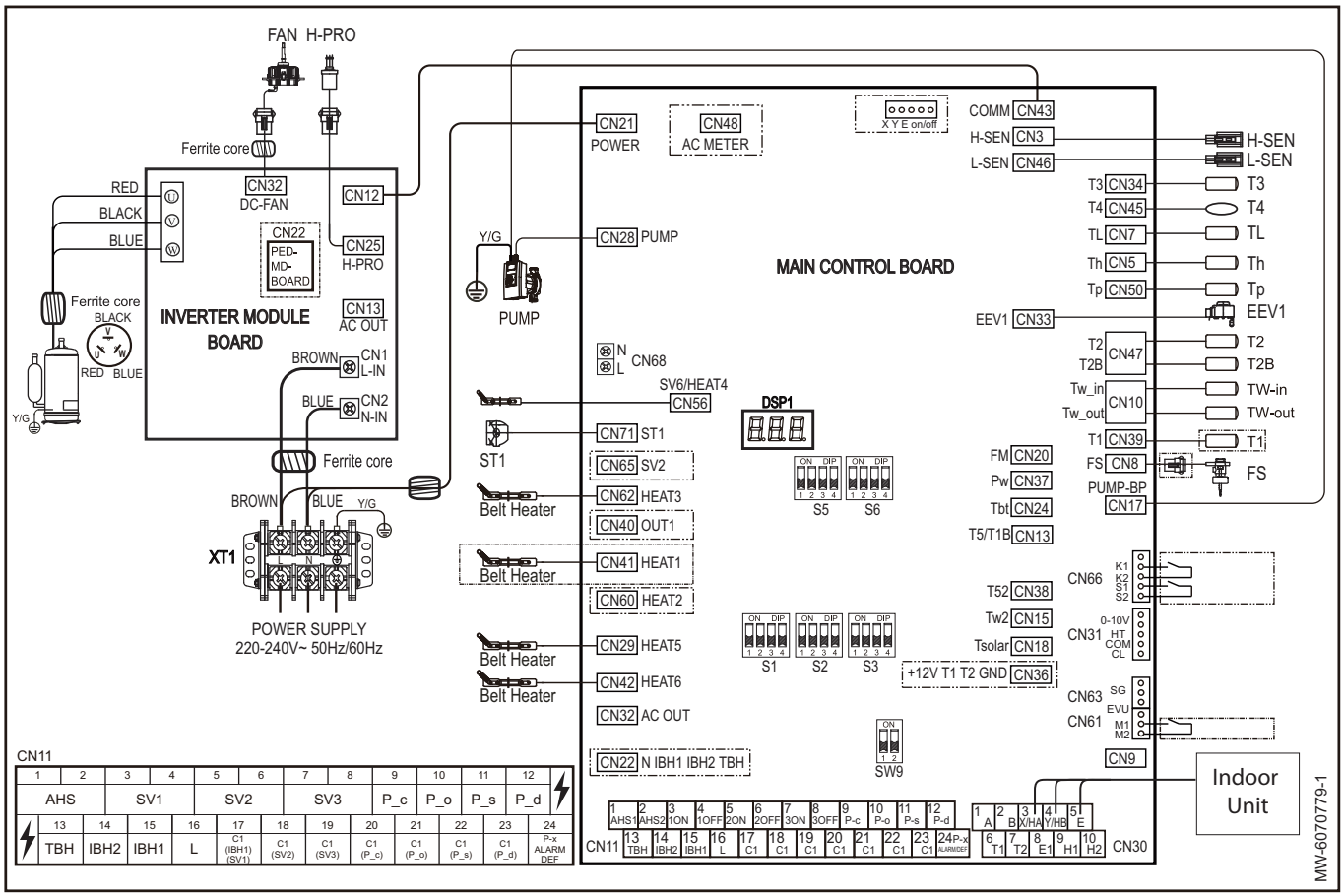
Fig.10 Assure HP55-8-1-PHMB - Assure HP55-11-1-PHMB - Assure HP55-11-3-PHMB - Assure HP55-13-1-PHMB - Assure HP55-13-3-PHMB



1 Compressor
2 Compressor discharge temperature sensor (high-pressure) Tp
3 Compressor intake temperature sensor (low pressure) Th
4 High-pressure switch
5 High-pressure sensor
6 4-way valve
7 Low pressure sensor
8 Outdoor air temperature sensor T4
9 Finned heat exchanger (evaporator in heating mode)
10 Fan
11 Heat exchanger temperature sensor T3
12 Heat exchanger outlet refrigerant temperature sensor (in cooling mode) TL
13 Filter
14 Electronic expansion valve
15 Refrigerant temperature sensor at the plate heat exchanger (in cooling mode) T2
16 Liquid receiver
17 Heating return temperature sensor Tw_in
18 Heat tape (plate heat exchanger)
19 Plate heat exchanger outlet refrigerant temperature sensor (in cooling mode) T2B
20 Plate heat exchanger (condenser in heating mode)
21 Heating flow temperature sensor Tw_out
22 Automatic bleed valve
23 Flow switch
24 Expansion vessel
25 Circulating pump
26 Safety pressure relief valve
Outlet Outlet - Heating circuit water flow
Inlet Inlet - Heating circuit water return
Cooling Cooling mode
Heating Heating mode

4.5 Electrical diagram

Fig.11 Single-phase outdoor units



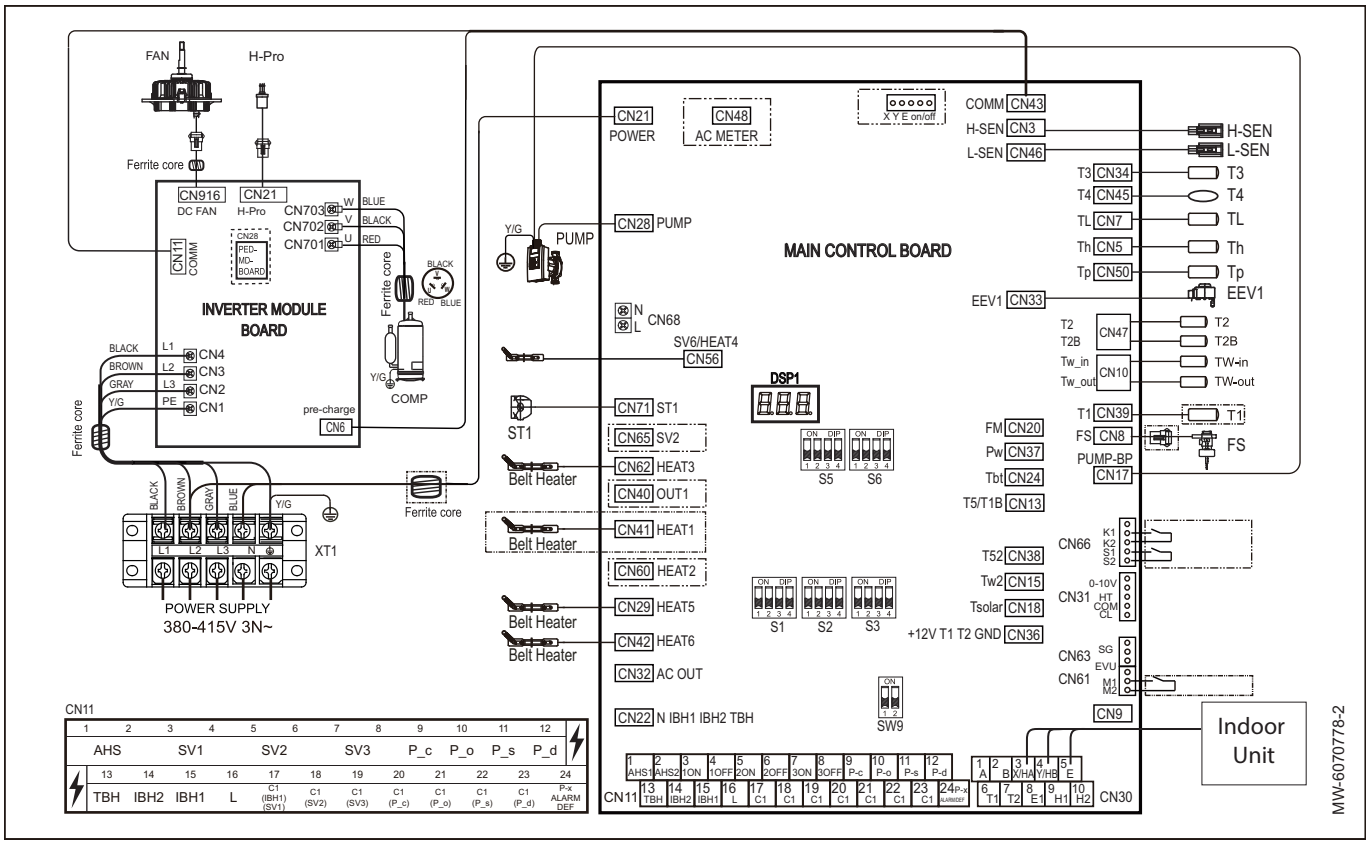
Tab.17

Item	Description
AC METER	Do not use
AC OUT	Transformer power input
AHS	Supplementary heat source
ALARM DEF	Do not use
Belt heater	Heating tape
BLACK	Black
BLUE	Blue
BROWN	Brown
COMM	Communication with the main control PCB
COM	Room thermostat
COMP	Inverter compressor
DSP1	Digital display
EEV1/2	Electric expansion valve
E1	Ground
EVU	Do not use
FAN	DC fan motor
Ferrite core	Ferrite core
FM	Flow rate sensor
FS	Flow switch
GND	Ground
H1, H2	Do not use
HEAT1/HEAT2	Do not use
HEAT3	Crankcase heater

Item	Description
HEAT5/HEAT6	Plate heat exchanger/Expansion vessel heat tape
H-Pro/L-PRO	High/low pressure switch
HT/CL	Heating mode/cooling mode (thermostat)
H-SEN/L-SEN	High/low pressure sensor
IBH1	Do not use
IBH2	Do not use
Indoor Unit	Indoor unit
INVERTER MODULE BOARD	Inverter module PCB
K1, K2	High-pressure switch
L	Live
N	Neutral
M1/M2	Remote switch
MAIN CONTROL BOARD	Main Control PCB
ON/OFF	On/Off
OUT1	Do not use
P_c	Do not use
P_d	Do not use
P_o	Do not use
P_s	Do not use
PED MD BOARD	PED PCB
POWER	Output
POWER SUPPLY	Power supply
PUMP	Pump
PUMP-BP	Variable speed pump communication
Pw	Water pressure sensor
RED	Red
S1, S2	High-pressure switch
SG	Do not use
ST1	Do not use
SV1	Do not use
SV2	Do not use
SV3	Do not use
SV6	Drainage outlet heating tape
T1	Water outlet temperature sensor downstream of the supplementary heater
T1B	Do not use
T2	Plate heat exchanger inlet refrigerant temperature sensor (for cooling mode)
T2B	Plate heat exchanger outlet refrigerant temperature sensor (for cooling mode)
T3	Heat exchanger temperature sensor
T4	Outdoor air temperature sensor
T5	Do not use
T52	Do not use
TBH	Control terminal board for cylinder backup device
Tbt	Do not use
TL	Heat exchanger outlet refrigerant temperature sensor (for cooling mode)
Th	Compressor suction temperature sensor
Tp	Compressor discharge temperature sensor
Tsolar	Tsolar
TW-in	Heating return temperature sensor
TW-out	Heating flow temperature sensor
Tw2	Do not use
XT1	Terminal

Item	Description
XYE	Bus connection
Y/G	Yellow/green

Fig.12 Three-phase outdoor units



Tab.18

Item	Description
AC METER	Do not use
AC OUT	Transformer power input
AHS	Supplementary heat source
ALARM DEF	Do not use
Belt heater	Heating tape
BLACK	Black
BLUE	Blue
BROWN	Brown
COMM	Communication with the main control PCB
COM	Room thermostat
COMP	Inverter compressor
DSP1	Digital display
EEV1/2	Electric expansion valve
E1	Ground
EVU	Do not use
FAN	DC fan motor
Ferrite core	Ferrite core
FM	Flow rate sensor
FS	Flow switch
GND	Ground
H1, H2	Do not use
HEAT1/HEAT2	Do not use
HEAT3	Crankcase heater

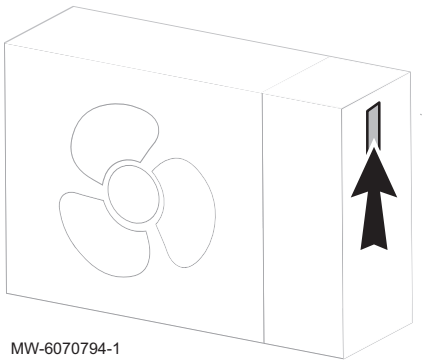
Item	Description
HEAT5/HEAT6	Plate heat exchanger/Expansion vessel heat tape
H-Pro/L-PRO	High/low pressure switch
HT/CL	Heating mode/cooling mode (thermostat)
H-SEN/L-SEN	High/low pressure sensor
IBH1	Do not use
IBH2	Do not use
Indoor Unit	Indoor unit
INVERTER MODULE BOARD	Inverter module PCB
K1, K2	High-pressure switch
L	Live
N	Neutral
M1/M2	Remote switch
MAIN CONTROL BOARD	Main Control PCB
ON/OFF	On/Off
OUT1	Do not use
P_c	Do not use
P_d	Do not use
P_o	Do not use
P_s	Do not use
PED MD BOARD	PED PCB
POWER	Output
POWER SUPPLY	Power supply
PUMP	Pump
PUMP-BP	Variable speed pump communication
Pw	Water pressure sensor
RED	Red
S1, S2	High-pressure switch
SG	Do not use
ST1	Do not use
SV1	Do not use
SV2	Do not use
SV3	Do not use
SV6	Drainage outlet heating tape
T1	Water outlet temperature sensor downstream of the supplementary heater
T1B	Do not use
T2	Plate heat exchanger inlet refrigerant temperature sensor (for cooling mode)
T2B	Plate heat exchanger outlet refrigerant temperature sensor (for cooling mode)
T3	Heat exchanger temperature sensor
T4	Outdoor air temperature sensor
T5	Do not use
T52	Do not use
TBH	Control terminal board for cylinder backup device
Tbt	Do not use
TL	Heat exchanger outlet refrigerant temperature sensor (for cooling mode)
Th	Compressor suction temperature sensor
Tp	Compressor discharge temperature sensor
Tsolar	Tsolar
TW-in	Heating return temperature sensor
TW-out	Heating flow temperature sensor
Tw2	Do not use
XT1	Terminal

Item	Description
XYE	Bus connection
Y/G	Yellow/green

5 Description of the product

5.1 Data plate

Fig.13



The data plate must be accessible at all times. It identifies the product and contains important information:

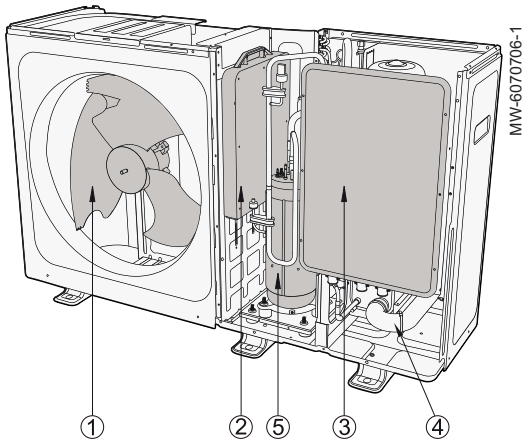
- the product type
- the date of manufacture (year - week)
- the serial number
- IP: the IP international protection marking
- Pe h: the electrical output of the heat pump (in nominal state)
- Pe bK: the electrical backup output
- Pth / COP the heat output of the heat pump/the performance coefficient (in nominal state)
- Ta: the outdoor air temperature limit
- Twh: the water temperature limit
- PSwh: the maximum water pressure
- Ref: the refrigerant type
- GWP / tCO_{2eq}: the Global Warming Potential/quantity of refrigerant in tonnes of CO_2 equivalent
- PS HP: the maximum pressure in normal operation
- Pmax: the maximum pressure at which the safety pressure relief valve is triggered

Never remove or cover the data plate and labels.

The data plate and labels must be legible throughout the entire lifetime of the product.

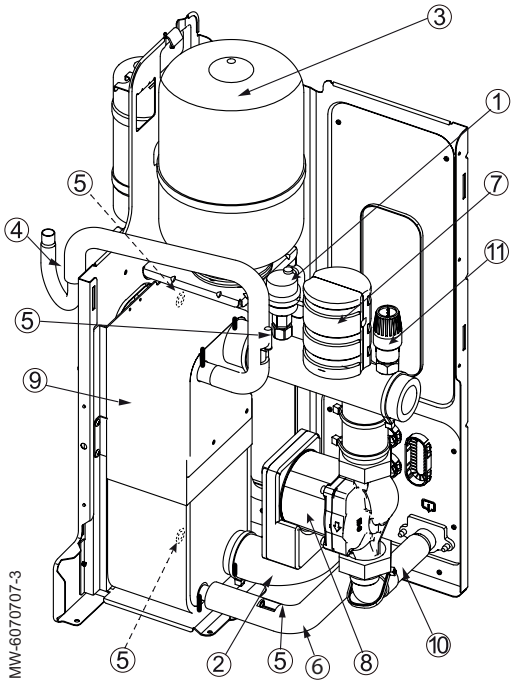
5.2 Main components of the outdoor unit

Fig.14



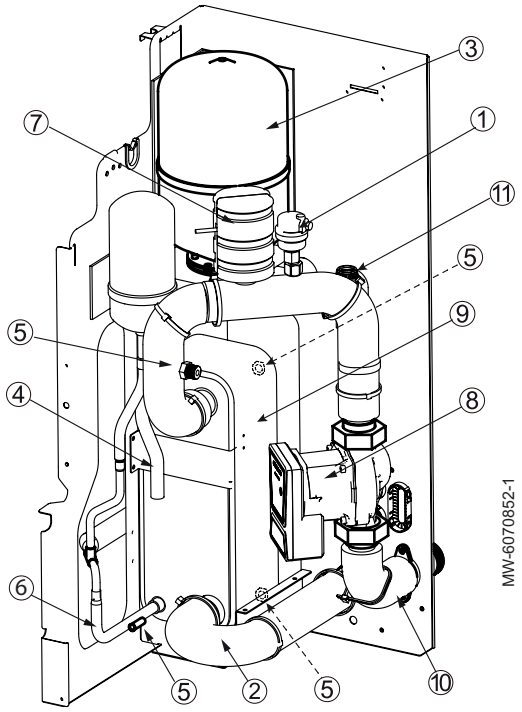
1 Fan
2 Inverter control box
3 Main control box
4 Hydraulic module
5 Compressor

Fig.15 Assure HP55-4-1-PHMB - Assure HP55-6-1-PHMB



1 Automatic bleed valve
2 Water inlet pipe
3 Expansion vessel
4 Refrigerant gas pipe
5 Temperature sensor
6 Refrigerant pipe
7 Flow switch
8 Circulating pump
9 Plate heat exchanger
10 Water outlet pipe
11 Safety pressure relief valve

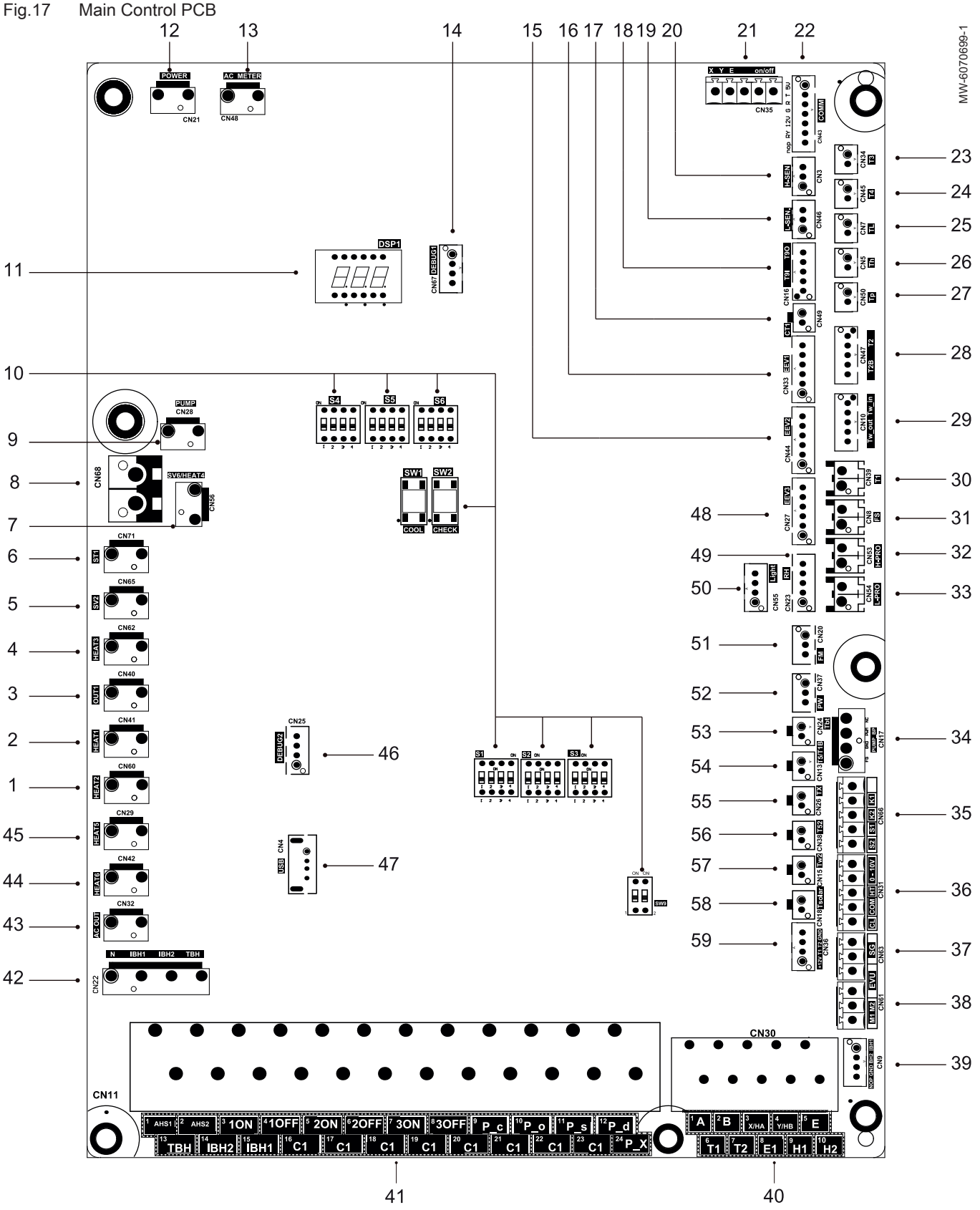
Fig.16 Assure HP55-8-1-PHMB - Assure HP55-11-1-PHMB - Assure HP55-11-3-PHMB - Assure HP55-13-1-PHMB - Assure HP55-13-3-PHMB



1 Automatic bleed valve
2 Water inlet pipe
3 Expansion vessel
4 Refrigerant gas pipe
5 Temperature sensor
6 Refrigerant pipe
7 Flow switch
8 Circulating pump
9 Plate heat exchanger
10 Water outlet pipe
11 Safety pressure relief valve

5.3 Terminal blocks

Fig.17 Main Control PCB



Tab.19

Number	Terminal block	Marking	Explanation	Terminal block voltage
1	CN60	HEAT2	Do not use	
2	CN41	HEAT1	Do not use	

Number	Terminal block	Marking	Explanation	Terminal block voltage
3	CN40	OUT1	Do not use	
4	CN62	HEAT3	Terminal block for crankcase heating tape	230 VAC
5	CN65	SV2	Do not use	
6	CN71	ST1	Do not use	230 VAC
7	CN56	/	Terminal block for the heating tape of bottom plate	230 VAC
8	CN68	/	Terminal block for the heating tape of drainage outlet	230 VAC
9	CN28	PUMP	Terminal block for variable speed pump power input	
10	/	/	Dip switch	
11	DSP1	/	Digital display	
12	CN21	POWER	Terminal block for power supply	230 VAC
13	CN48	AC METER	Do not use	
14	CN67	DEBUG1	Terminal block for printed circuit programming	
15	CN44	EEV2	Terminal block for electrical expansion valve 2	
16	CN33	EEV1	Do not use	
17	CN49	CT1	Do not use	
18	CN16	T9O/T9I	Do not use	
19	CN46	L-SEN	Terminal block for low pressure sensor	0-5 VDC
20	CN3	H-SEN	Terminal block for high pressure sensor	0-5 VDC
21	CN35	RS485	Do not use	
		on/off	Do not use	
22	CN43	COMM	Terminal block for communication with Inverter module	0-5 VDC
23	CN34	T3	Terminal block for T3 temperature sensor	0-3.3 V DC
24	CN45	T4	Terminal block for T4 temperature sensor	0-3.3 V DC
25	CN7	TL	Terminal block for TL temperature sensor	0-3.3 V DC
26	CN5	Th	Terminal block for Th temperature sensor	0-3.3 V DC
27	CN50	Tp	Terminal block for Tp temperature sensor	0-3.3 V DC
28	CN47	T2	Terminal block for refrigerant fluid side temperature (heating mode)	0-5 VDC
		T2B	Terminal block for temperature sensors of refrigerant gas side temperature (cooling mode)	0-5 VDC
29	CN10	TW_in	Terminal block for plate heat exchanger water inlet temperature sensors	0-5 VDC
		TW_out	Terminal block for plate heat exchanger water outlet temperature sensors	0-5 VDC
30	CN39	T1	Terminal block for final water outlet temperature sensors	0-5 VDC
31	CN8	FS	Terminal block for the flow switch	0-12 VDC
32	CN53	H-PRO	Terminal block for high pressure switch	
33	CN54	L-PRO	Terminal block for low pressure switch	
34	CN17	PUMP_BP	Terminal block for variable speed pump communication	0-5 VDC
35	CN66	K1,K2	Terminal block for high pressure switch	0-5 VDC
		S1,S2	Terminal block for high pressure switch	0-5 VDC
36	CN31	0~10 V	Output port for 0-10 V	0-5 VDC
		HT	Control terminal board for the room thermostat (heating mode)	0-5 VDC
		COM	Power terminal block for room thermostat	0-5 VDC
		CL	Control terminal board for the room thermostat (cooling mode)	0-5 VDC
37	CN63	SG	Do not use	0-12 VDC
		EVU	Do not use	0-12 VDC
38	CN61	M1 M2	Terminal block for remote switch	0-12 VDC
39	CN9	/	Do not use	

Number	Terminal block	Marking	Explanation	Terminal block voltage
40	CN30	1.2	Terminal block for supplementary heat source	
		3.4	Terminal block for communication with the wired controller	
		5	Do not use	
		6.7	Terminal block for thermostat transfer board	
		8	Do not use	
		9.10	Terminal block for machine cascade	
41	CN11	1, 2	Terminal block for supplementary heat source	230 VAC
		3, 4, 17	Do not use	230 VAC
		5, 6, 18	Do not use	230 VAC
		7, 8, 19	Do not use	230 VAC
		9, 20	Do not use	230 VAC
		10, 21	Do not use	230 VAC
		11, 22	Do not use	230 VAC
		12, 23	Do not use	230 VAC
		13, 16	Do not use	230 VAC
		14, 16	Do not use	230 VAC
		15, 17	Do not use	230 VAC
		24	Do not use	230 VAC
42	CN22	IBH1	Do not use	230 VAC
		IBH2	Do not use	230 VAC
		TBH	Control terminal board for cylinder supplementary device	230 VAC
43	CN32	AC OUT	Terminal block for transformer power supply	230 VAC
44	CN42	HEAT6	Terminal block for internal anti-freeze electric heating tape	230 VAC
45	CN29	HEAT5	Terminal block for internal anti-freeze electric heating tape	230 VAC
46	CN25	DEBUG2	Terminal block for printed circuit programming	
47	CN4	USB	Terminal block for USB programming	
48	CN27	EEV3	Terminal block for electrical expansion valve	
49	CN23	RH	Terminal block for humidity sensor	
50	CN55	Light	Terminal block for indicator light	
51	CN20	FM	Terminal block for water flow rate sensor	0-5 VDC
52	CN37	PW	Terminal block for water pressure sensor	0-5 VDC
53	CN24	Tbt	Do not use	0-5 VDC
54	CN13	T5/T1B	Do not use	0-5 VDC
55	CN26	TX	Do not use	
56	CN38	T52	Do not use	0-5 VDC
57	CN15	Tw2	Do not use	0-5 VDC
58	CN18	Tsolar	Do not use	0-5 VDC
59	CN36	/	Terminal block for thermostat transfer board	0-12 VDC

6 Installation

6.1 Checking the condition of the packaging of the outdoor unit

The outdoor unit contains a highly flammable refrigerant. On receipt of the outdoor unit, you need to make sure that it has not sustained any impact which might have caused a refrigerant leak.

Danger
If the packaging shows any sign of damage or impact, do not install the outdoor unit.
Take the following measures to prevent any risk associated with a refrigerant leak:

1. Immediately move the outdoor unit in the open air, at least 6 metres away from any ignition source.
2. When moving and storing, keep any ignition sources away, especially motorised or electrical appliances, phones and cigarettes.
3. Contact the After-Sales service for the next step in the procedure.

If there is a refrigerant leak, it will dissipate into the atmosphere after a few hours. After four hours, use a sniffer to check that there is no refrigerant at all.

6.2 Selecting the location of the outdoor unit

Install the outdoor unit outdoors exclusively. The location of the outdoor unit must meet the safety, accessibility, comfort and performance recommendations.

1. Take account of the following parameters when choosing a location:
   - The protective perimeter around the outdoor unit
   - The minimum distances from the walls
   - The climate conditions
   - Noise pollution
   - The maximum distance from the indoor unit
   - The legal requirements

See
Indoor unit manual

6.2.1 Protective perimeter

The outdoor unit contains R290 refrigerant. This highly flammable refrigerant is denser than air and can accumulate at ground level in the event of a leak.

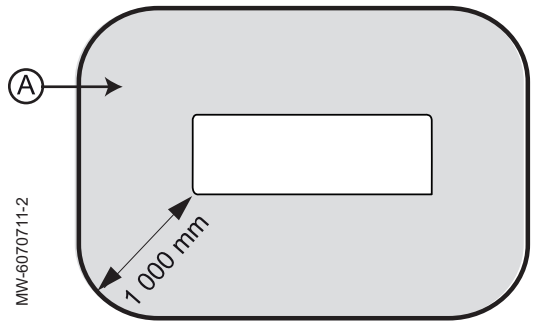
A protective perimeter must be marked out around the outdoor unit to ensure the safety of users in the event of a leak:

- It must not be possible for refrigerant to accumulate so as to form a toxic, suffocating, explosive or dangerous atmosphere.
- The refrigerant must not penetrate into buildings via openings, or into the waste water network.
- It must not be possible for refrigerant to accumulate inside cavities.

The protective perimeter applies to all types of installation, including wall mounting, and extends to the ground.

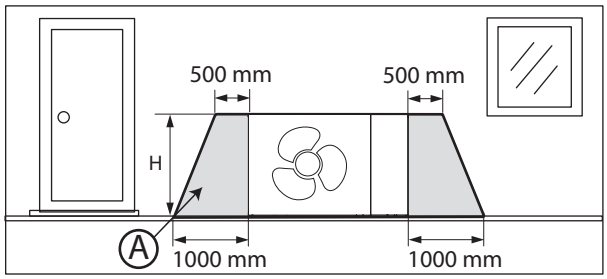
- The protective perimeter must not include any openings to the inside of the home: doors, windows, skylights, dome windows, air vents, lightwells, cellar access or sewer hatches.
- The protective perimeter must be free from permanent ignition sources such as light bulbs, electrical switches, electrical sockets or any other permanent ignition sources.
- The protective perimeter must be free from temporary sources of ignition such as lawnmowers, barbecues, cigarettes or any other temporary ignition sources.
- The protective perimeter must not adjoin any alleys, parking spaces, neighbouring plots or public areas.

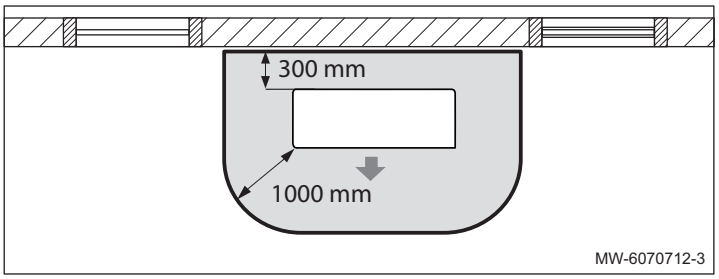
Fig.18 Ground-mounted, on a plot or on a flat roof



A Protective perimeter

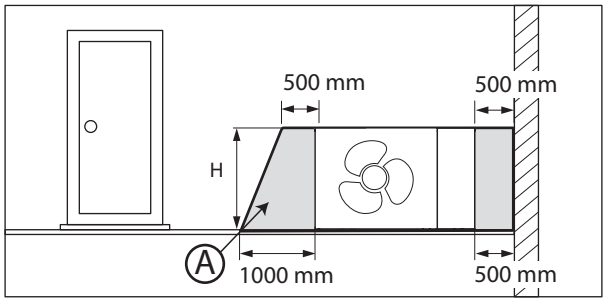
Fig.19 Siting the outdoor unit in front of an external wall

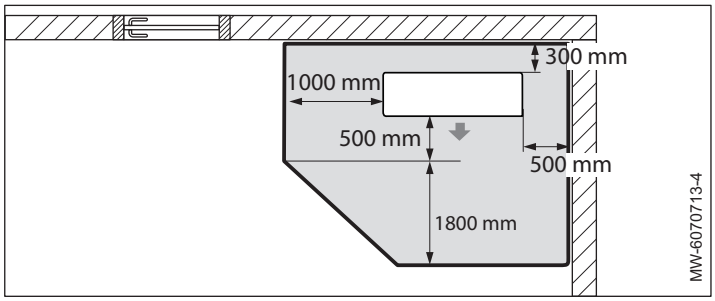




A Protective perimeter
H Height of the protection perimeter: greater than or equal to the height of the outdoor unit in relation to the ground

Fig.20 Installing the outdoor unit in a corner





A Protective perimeter
H Height of the protection perimeter: greater than or equal to the height of the outdoor unit in relation to the ground

6.2.2 Minimum distances from the walls

The outdoor unit must be installed away from walls to ensure its proper operation, accessibility during maintenance work, and the safety of property and people.

Nothing must obstruct the free circulation of air around the outdoor unit (intake and supply air).

Fig.21 With obstacle on the top

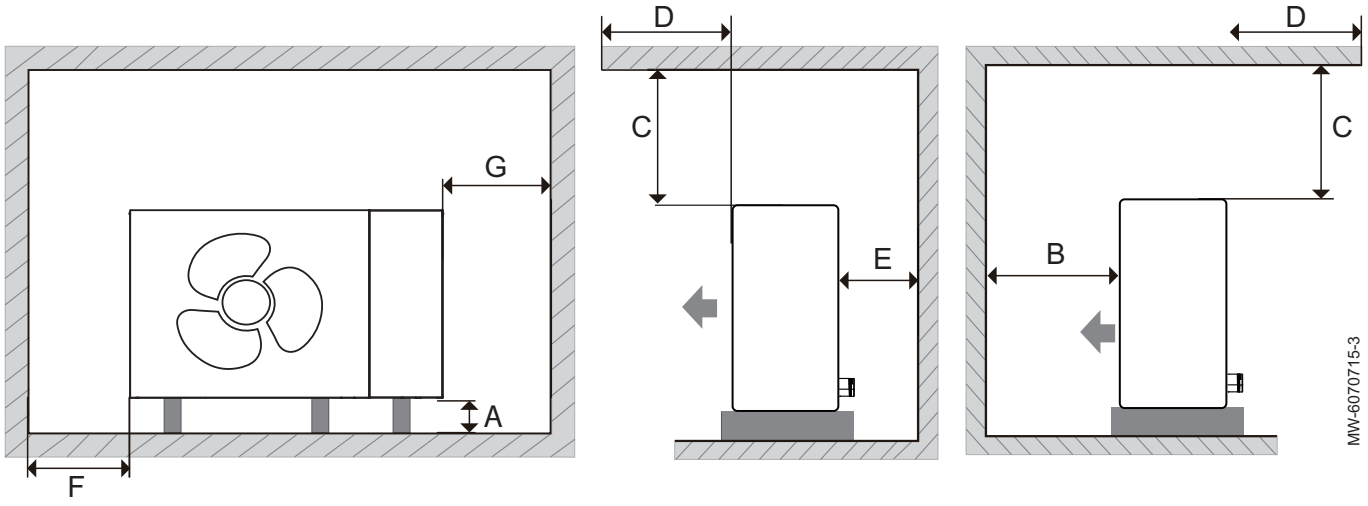
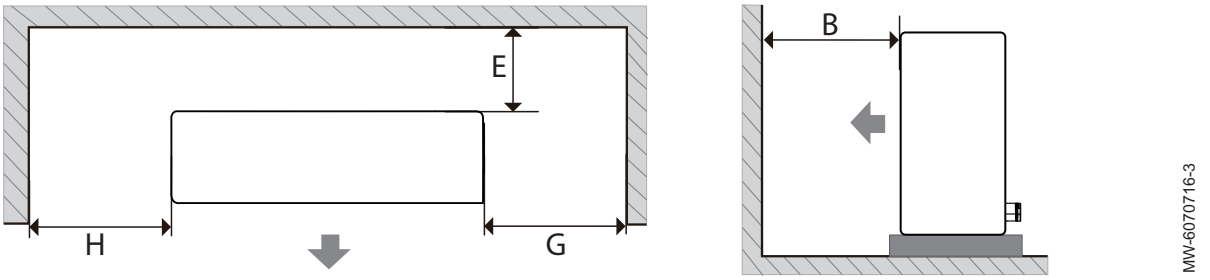


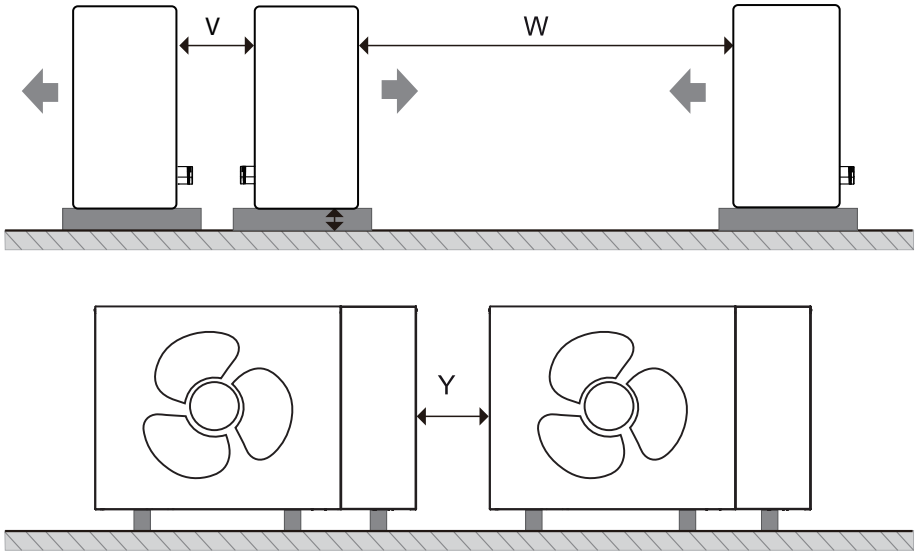
Fig.22 Without obstacle on the top



Outdoor unit	Unit	A(1)	B	C	D	E	F	G	H
Assure HP55-4-1-PHMB	mm	≥ 100	≥ 1000	≥ 500	≤ 500	≥ 300	≥ 500	≥ 500	≥ 500
Assure HP55-6-1-PHMB	mm	≥ 100	≥ 1000	≥ 500	≤ 500	≥ 300	≥ 500	≥ 500	≥ 500
Assure HP55-8-1-PHMB	mm	≥ 100	≥ 1000	≥ 500	≤ 500	≥ 300	≥ 500	≥ 500	≥ 500
Assure HP55-11-1-PHMB	mm	≥ 100	≥ 1500	≥ 500	≤ 500	≥ 300	≥ 500	≥ 500	≥ 500
Assure HP55-11-3-PHMB	mm	≥ 100	≥ 1500	≥ 500	≤ 500	≥ 300	≥ 500	≥ 500	≥ 500
Assure HP55-13-1-PHMB	mm	≥ 100	≥ 1500	≥ 500	≤ 500	≥ 300	≥ 500	≥ 500	≥ 500
Assure HP55-13-3-PHMB	mm	≥ 100	≥ 1500	≥ 500	≤ 500	≥ 300	≥ 500	≥ 500	≥ 500

(1) In cold weather, take account of snow on the ground.

Fig.23 Clearance between outdoor units



Outdoor unit	Unit	V	W	Y
Assure HP55-4-1-PHMB	mm	≥ 600	≥ 2500	≥ 500
Assure HP55-6-1-PHMB	mm	≥ 600	≥ 2500	≥ 500
Assure HP55-8-1-PHMB	mm	≥ 600	≥ 2500	≥ 500
Assure HP55-11-1-PHMB	mm	≥ 600	≥ 3000	≥ 500
Assure HP55-11-3-PHMB	mm	≥ 600	≥ 3000	≥ 500
Assure HP55-13-1-PHMB	mm	≥ 600	≥ 3000	≥ 500
Assure HP55-13-3-PHMB	mm	≥ 600	≥ 3000	≥ 500

For the clearance in other directions, refer to the preceding diagrams.

See also
Installation in cold climates, page 34

6.2.3 General rules

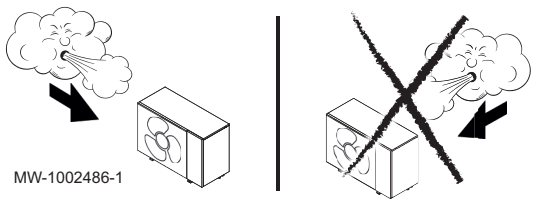
In addition to the "protection perimeter", the following conditions must be met:

- keep the outdoor unit away from flammable substances or flammable gases.
- do not expose the outdoor unit to a dirty, dusty or corrosive atmosphere.
- protect the outdoor unit from climbing plants and leaves that could block the air intake and outlet of the outdoor unit.
- make sure that no insects, snakes or small animals get into the outdoor unit.
- regularly check that no wild animals have damaged the piping and wiring of the outdoor unit.

Important
In case of any evidence of animal effects, ask professionals for inspection and maintenance.

6.2.4 Exposure to strong winds

Fig.24



When installing the outdoor unit in a place exposed to strong wind, pay special attention to the following:

A wind speed of 5 m/s or higher against the outdoor unit's air outlet could cause a short circuit (suction of discharge air), which may have the following consequences:

- deterioration of the operational capacity.
- frequent frosting in heating mode.
- disruption of operation due to a pressure rise.

When a continuous strong wind blows against the front of the outdoor unit, there is a risk of the fan running at excessive speed and breaking down.

The air inlet of the outdoor unit must be protected from prevailing winds. If the installation's location prevents this:

- position the outdoor unit so as to minimise wind resistance
- or install a protective wall behind the outdoor unit. In this case, adhere to the recommended wall distances.

6.2.5 Installation in coastal areas

In coastal areas, the outdoor unit must be protected from corrosion and sea breezes.
Corrosion may shorten the service life and affect the performance of the outdoor unit.

Do not expose the outdoor unit directly to sea breezes.

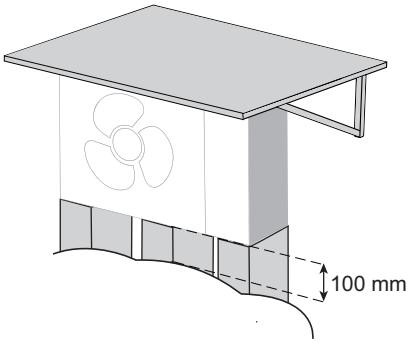
Do not install the outdoor unit in immediate proximity to the sea. A minimum distance of 500 metres is required.

6.2.6 Exposure to strong solar radiation

Installing the outdoor unit in direct sunlight can affect the measurement of the outdoor temperature and have an impact on the performance of the outdoor unit. The colour of the outdoor unit can also fade over time. Shade the outdoor unit.

6.2.7 Installation in cold climates

Fig.25

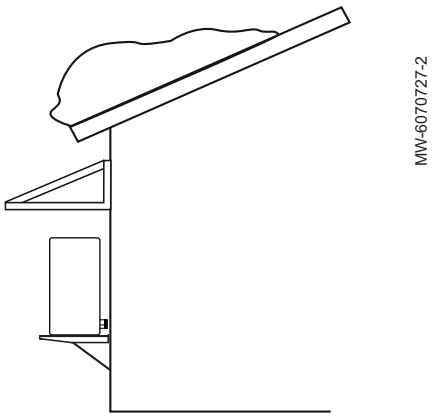


It is recommended that the outdoor unit be placed with the back side oriented towards the wall.

Install a canopy over the outdoor unit to prevent snow from accumulating on the outdoor unit in extreme weather conditions.

Install a high pedestal or wall mount the outdoor unit to keep a proper clearance (at least 100 mm) between the outdoor unit and the mean snow cover.

Fig.26



If there is a risk of snow slip from the roof, a protective roof or cover must be erected to protect the heat pump, pipes and wiring.

6.2.8 Managing acoustic comfort

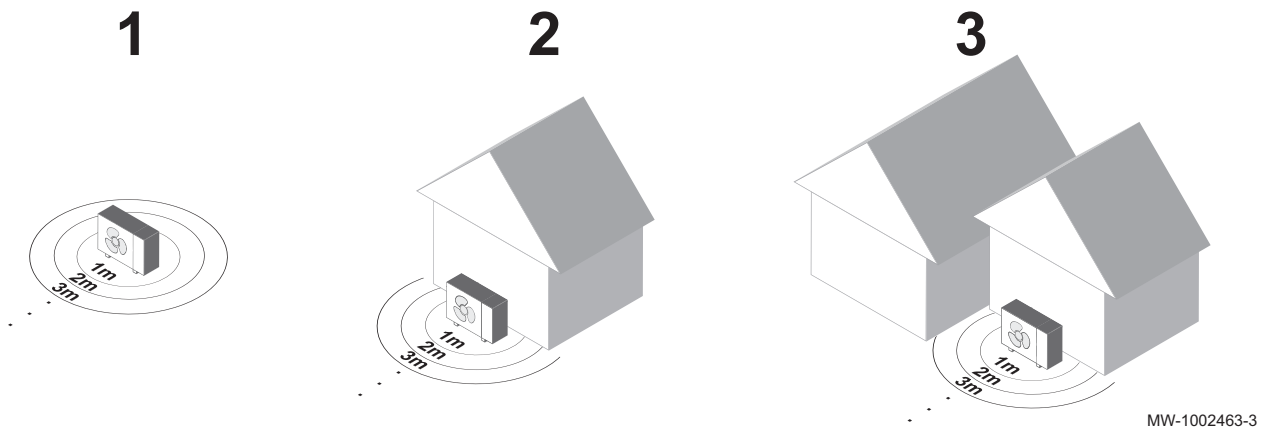
The choice of the location for the outdoor unit must take acoustic comfort into account.

- Install the outdoor unit away from sleeping areas and terraces
- Do not install the outdoor unit facing a glass wall

The outdoor unit is fitted with a non-rigid connection to prevent vibrations being transmitted to the building, thereby maximising acoustic comfort.

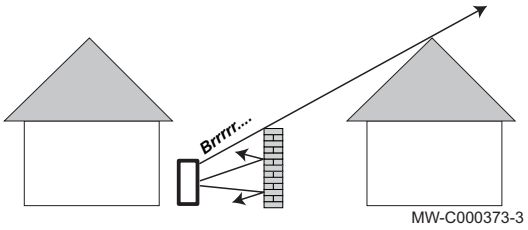
The noise level varies depending on the distance to the outdoor unit and on the mounting type selected.

Fig.27



1 Mounting on a plot or flat roof
2 Mounting in front of a building wall
3 Mounting in the corner of a building

Fig.28



When the outdoor unit is too close to neighbours, install a noise reduction screen to reduce noise pollution. Install the noise reduction screen as close as possible to the source of noise.

This type of equipment must be installed in compliance with:

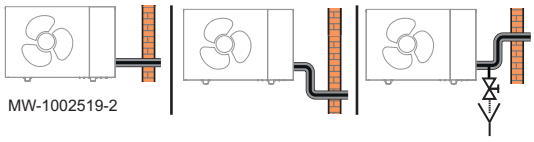
- The legislation
- Prevailing standards
- The minimum positioning distances in relation to the outdoor unit
- Free air circulation in the outdoor unit exchanger
- The accessibility of the outdoor unit for maintenance interventions

6.3 Preparing for installation

1. Choose the assembly type:
   - ground assembly on rubber mounts
   - ground assembly on crossbar and rubber mounts
   - wall assembly
   - flat roof assembly

Important
Installation on a sloping surface is not permitted. The outdoor unit must be installed horizontally.

Fig.29



2. Prepare a gravel bed for the condensate drain.
3. Prepare the location for the outdoor unit and its bracket:
   - The outdoor unit must be placed on a flat, stable surface capable of bearing its weight
   - The configuration of the water connections must enable the outdoor unit to be drained (drain valve at low point)

Important
Beware of the risk of freezing if the valve is installed outdoors.

4. Put the bracket in place.

See
Manual for the chosen support

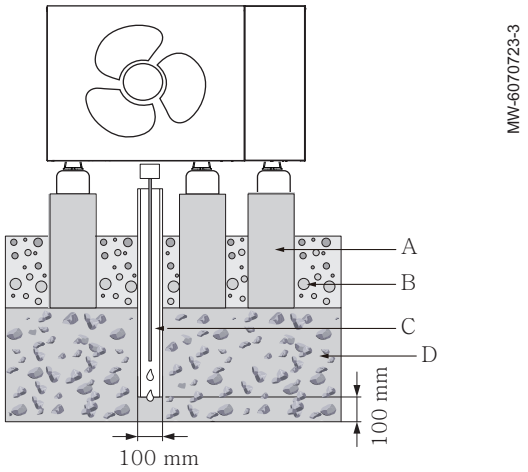
6.3.1 Condensate drainage

Make sure that any condensate that accumulates is drained off in a frost-free area.

To prevent the condensate from freezing, a self-regulating heating cable (not supplied) can be threaded into the downpipe so that the condensate can drain via the downpipe.

Open gullies within the protection perimeter do not pose any safety risk.

Fig.30 Condensate drain into a gravel bed

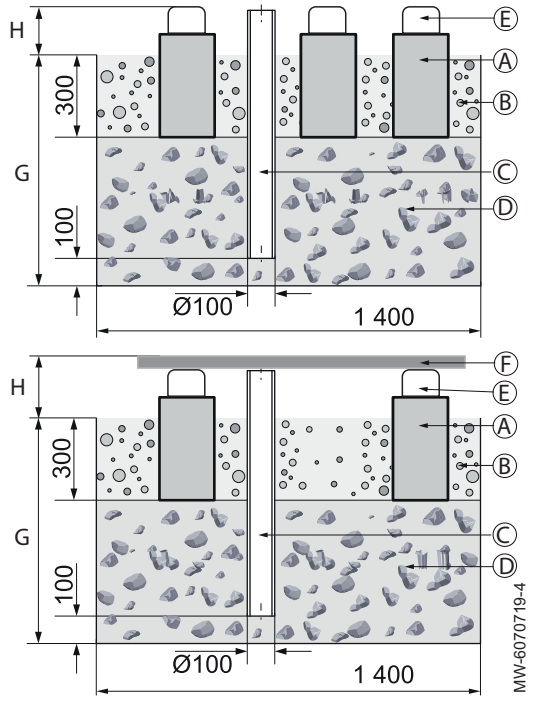


A Concrete blocks
B Foundation
C Discharge pipe
D Gravel bed

The downpipe must go into a sufficiently large gravel bed so that the condensate can drain away freely.

See also
Preparing a ground installation, page 36

6.3.2 Preparing a ground installation

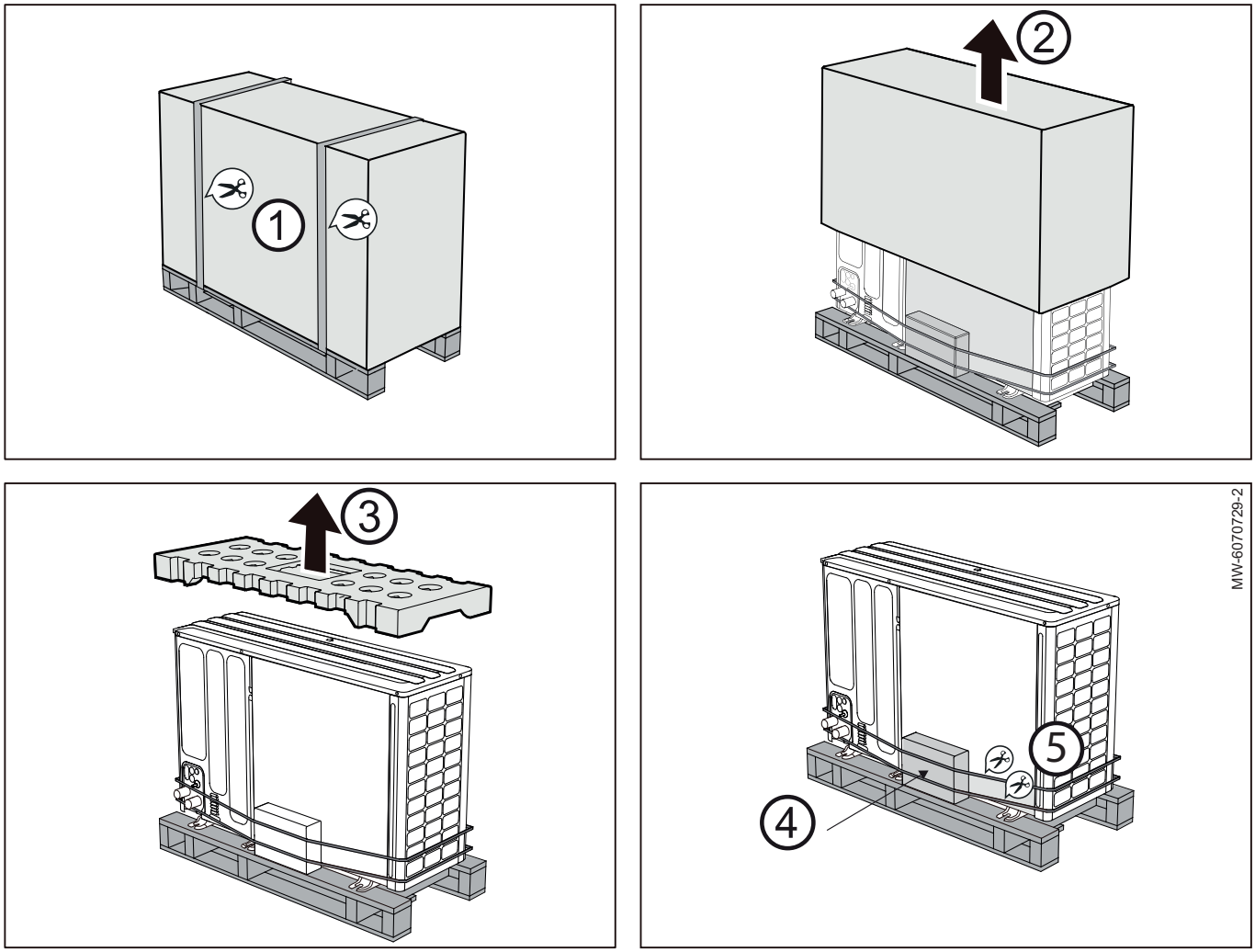


A Concrete blocks
B Foundations
C Downpipe for the condensate drain
D Water-permeable coarse rubble
E Rubber mounts (accessory)
F Crossbar (accessory)
G Minimum depth: 900 mm where the ground freezes; 600 mm where the ground does not freeze
H Foundation height to suit local conditions. Such height must not be smaller than 100 mm.

1. Dig a hole in the ground.
2. Insert a downpipe to divert the condensate.
3. Add a layer of water-permeable coarse rubble.
4. Create concrete blocks adapted to the chosen accessory for installation on the ground
5. Add a bed of gravel between the foundations strips to divert condensate.

6.4 Unpack the outdoor unit

Fig.31



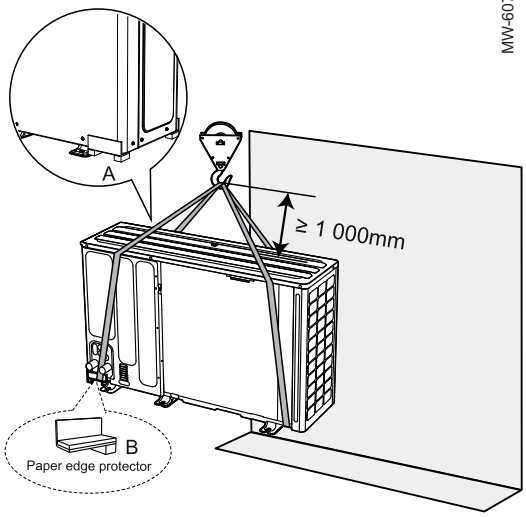
1. Cut the strapping.
2. Remove the cardboard.
3. Remove the protective cover.
4. Check that the accessory box is present.
5. Cut the strapping.

6.5 Transport

6.5.1 Transporting the outdoor unit using a crane

The outdoor unit is a large-size appliance. Transporting it with a crane requires a few precautions.

Fig.32



A Paper edge protector A
B Paper edge protector B

Caution
The barycentre of the product and the hook should be kept vertically aligned to prevent excessive tilting.

1. Use lifting tools with transport straps or a suitable pallet truck.

Location of the outdoor unit	Action
Outdoor unit on the pallet	Pass the transport straps through the holes on the left and right sides of the pallet correctly.
No pallet under the outdoor unit	The transport straps can be fitted into the recesses in the base frame that are made specifically for this purpose. Use a paper edge protector under the outdoor unit when lifting the outdoor unit.

2. Check that the equipment has not been damaged during transportation or storage.
3. In the event of damage, take the necessary precautions to avoid the risk of refrigerant leakage and contact the after-sales service.

See also
Dimensions and connections, page 16

6.5.2 Transporting the outdoor unit without a crane

The outdoor unit is a heavy and large-size appliance. Transporting it by hand requires a few precautions.

Four people are required to lift and move the outdoor unit.

1. Take into consideration the weight distribution during transportation. The outdoor unit is significantly heavier on the compressor side than on the fan motor side.

Important
During transport, do not tilt the outdoor unit significantly to prevent it from tipping over.

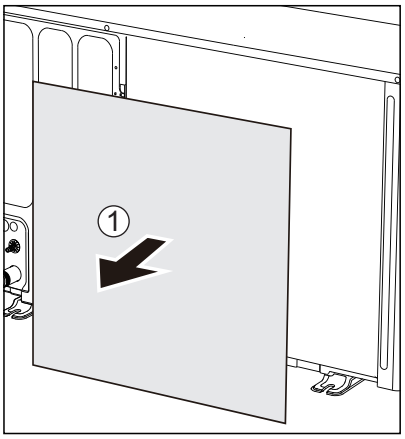
2. Protect the casing sections from damage. Use a paper edge protector under the outdoor unit when lifting the outdoor unit.
3. Check that the equipment has not been damaged during transportation or storage.
4. In the event of damage, take the necessary precautions to avoid the risk of refrigerant leakage and contact the after-sales service.

See also
Dimensions and connections, page 16

6.6 Putting the outdoor unit in place

6.6.1 Removing the exchanger protection

Fig.33



1. Remove the exchanger protection.

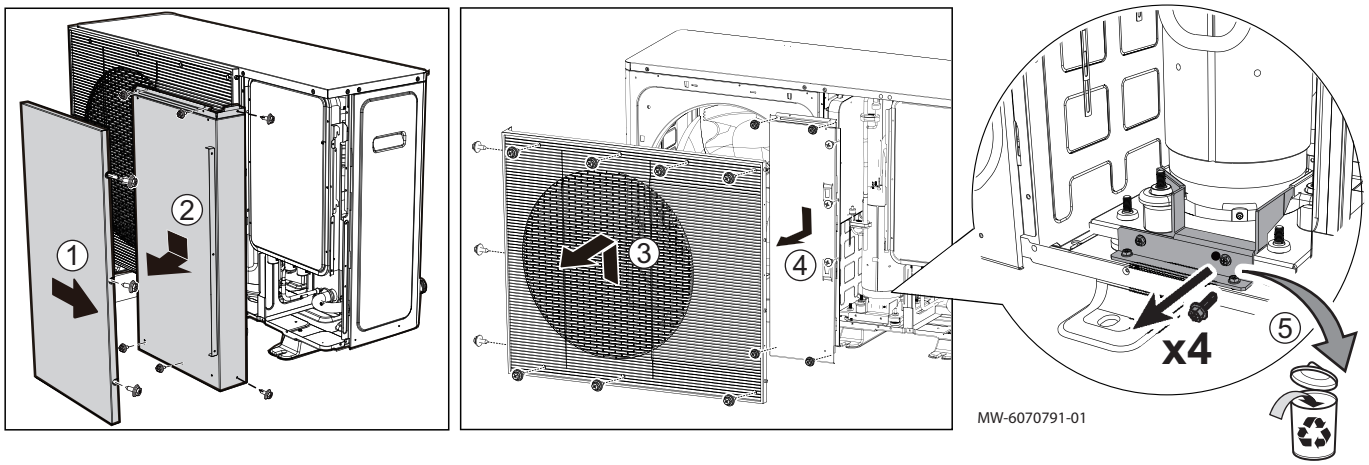
Important
Take care not to damage the outdoor temperature sensor.

2. Check the condition of the exchanger.

6.6.2 Remove the compressor support from the outdoor units Assure HP55-8-1-PHMB - Assure HP55-11-1-PHMB - Assure HP55-13-1-PHMB - Assure HP55-11-3-PHMB - Assure HP55-13-3-PHMB

The compressor support holds the refrigeration module in place during transport. It must be removed once the outdoor unit has been installed.

Fig.34

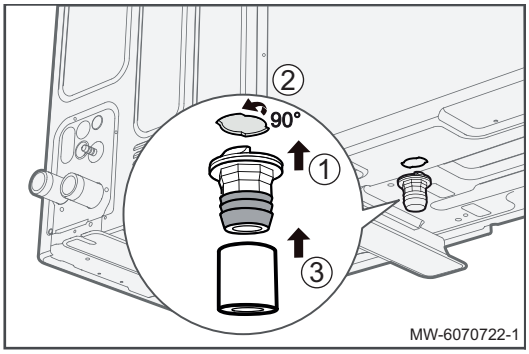


1. Remove the front panel.
2. Remove the inner panel.
3. Remove the grille.
4. Remove the compressor compartment cover.
5. Remove the compressor support.

Important
Failure to remove the compressor support will invalidate the warranty.

6.6.3 Connecting the condensate drain hose

Fig.35



1. Insert the condensate drain union (supplied in the accessories box) into the special hole.
2. Turn the union to secure it in its housing.
3. Connect the condensate drain hose to the union.
4. Connect the condensate drain hose outlet to the waste water discharge.
5. Make sure the condensate is drained properly. If not, use the additional drain hole.

See also
Dimensions and connections, page 16

6.7 Hydraulic connections

6.7.1 Minimum volume of water

The volume of water in the installation must be sufficient to avoid short cycle operation and to enable optimal defrosting.

Important
The minimum volume of water in circulation must be available at any time, even if there is no heating demand or when all the valves are closed.

If the volume of water in circulation is not sufficient, it is necessary to install a buffer tank with the additional volume.

Tab.20 35 °C application - underfloor heating

	Assure HP55-4-1-PHMB	Assure HP55-6-1-PHMB	Assure HP55-8-1-PHMB	Assure HP55-11-1-PHMB	Assure HP55-11-3-PHMB	Assure HP55-13-1-PHMB	Assure HP55-13-3-PHMB
Minimum volume of water in circulation (l)	27	29	77	81	81	91	91
Minimum volume of glycol water in circulation (l)	31	34	89	94	94	105	105

Tab.21 45°C application - low temperature radiators or convection fans

	Assure HP55-4-1-PHMB	Assure HP55-6-1-PHMB	Assure HP55-8-1-PHMB	Assure HP55-11-1-PHMB	Assure HP55-11-3-PHMB	Assure HP55-13-1-PHMB	Assure HP55-13-3-PHMB
Minimum volume of water in circulation (l)	23	23	49	54	54	59	59
Minimum volume of glycol water in circulation (l)	27	27	57	63	63	68	68

Tab.22 55 °C application - low temperature radiators

	Assure HP55-4-1-PHMB	Assure HP55-6-1-PHMB	Assure HP55-8-1-PHMB	Assure HP55-11-1-PHMB	Assure HP55-11-3-PHMB	Assure HP55-13-1-PHMB	Assure HP55-13-3-PHMB
Minimum volume of water in circulation (l)	26	26	42	49	49	51	51
Minimum volume of glycol water in circulation (l)	30	30	49	57	57	59	59

Tab.23 65 °C application - high temperature radiators

	Assure HP55-4-1-PHMB	Assure HP55-6-1-PHMB	Assure HP55-8-1-PHMB	Assure HP55-11-1-PHMB	Assure HP55-11-3-PHMB	Assure HP55-13-1-PHMB	Assure HP55-13-3-PHMB
Minimum volume of water in circulation (l)	26	26	38	49	49	49	49
Minimum volume of glycol water in circulation (l)	30	30	44	57	57	57	57

6.7.2 Heating water specifications

Before filling the heating installation, the water quality must be checked:

- The water must be clear and free of sedimentary substances and must not contain foreign matter such as welding beads, rust particles, scale, mud or other sedimentary substances
- The quality of the filling water must comply with the specifications given in the table below:

Tab.24 Heating water specifications

Specification	Value
Potential of hydrogen (pH)	7.5 - 9
Conductivity at 20 °C	< 500 µS/cm
Chlorides	Below 50 mg/l
Other components	Below 1 mg/l
Total hardness	20 °fH
	11.2 °dH
	2.0 mmol/l

Caution
Do not add any chemical products to the heating water without first consulting a water treatment specialist. For example: antifreeze, water softeners, products to increase or reduce the pH value, chemical additives and/or inhibitors. These may cause faults in the heat pump and damage the heat exchanger.

Warranty claims are excluded if the specified values are not adhered to or if any documentation is missing.

See also
Filling the heating circuit with mains water, page 52
Filling the heating circuit with glycol water, page 53

6.7.3 Volume of the expansion vessel

The volume of the expansion vessel must be compatible with the volume of water in the circuit taking into consideration the maximum temperature in heating mode.

If the volume of the expansion vessel integrated in the outdoor unit (5 effective litres) is not sufficient, add an external vessel to the heating circuit.

Tab.25 Underfloor heating type installation: maximum temperature of 40 °C

Static height	Expansion vessel inflation pressure	Volume of the installation (l)							
		75	100	125	150	175	200	225	250
		Expansion vessel volume (l)							
5 m	0.1 MPa (1 bar)	7	7	8	8	8	9	9	9
10 m	0.13 MPa (1.3 bar)	7	8	8	9	9	10	10	11
15 m	0.18 MPa (1.8 bar)	10	10	11	11	12	13	13	14

Tab.26 Radiator type installation: maximum temperature of 70 °C

Static height	Expansion vessel inflation pressure	Volume of the installation (l)							
		75	100	125	150	175	200	225	250
		Expansion vessel volume (l)							
5 m	0.1 MPa (1 bar)	8	9	10	11	12	13	14	15
10 m	0.13 MPa (1.3 bar)	9	11	12	13	14	15	16	17
15 m	0.18 MPa (1.8 bar)	12	13	15	16	18	19	21	22

6.7.4 Flow rate range

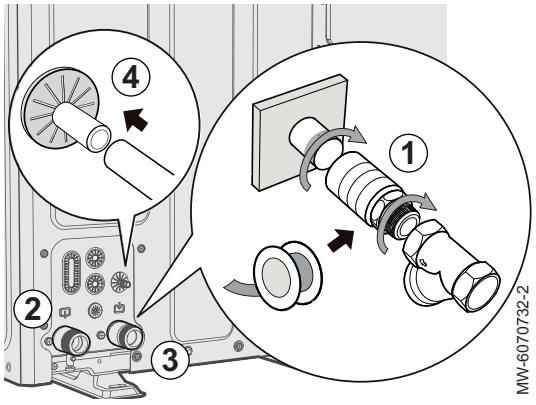
Check that the minimum flow rate in the installation is guaranteed in all conditions. This rate is required during defrosting/supplementary heater operation.

When one or more heating circuits are controlled by remotely controlled valves, the minimum water flow needs to be guaranteed, even if all valves are closed. If the minimum flow rate cannot be satisfied, E0 and E8 (outdoor unit shutdown) will be triggered.

Outdoor unit	Flow rate range
Assure HP55-4-1-PHMB	6.67 to 15 l/min
Assure HP55-6-1-PHMB	6.67 to 20.83 l/min
Assure HP55-8-1-PHMB	6.67 to 35 l/min
Assure HP55-11-1-PHMB Assure HP55-11-3-PHMB	11.67 to 41.67 l/min
Assure HP55-13-1-PHMB Assure HP55-13-3-PHMB	11.67 to 50 l/min

6.7.5 Connecting the outdoor unit to the hydraulic circuit

Fig.36 Assure HP55-4-1-PHMB - Assure HP55-6-1-PHMB



Important

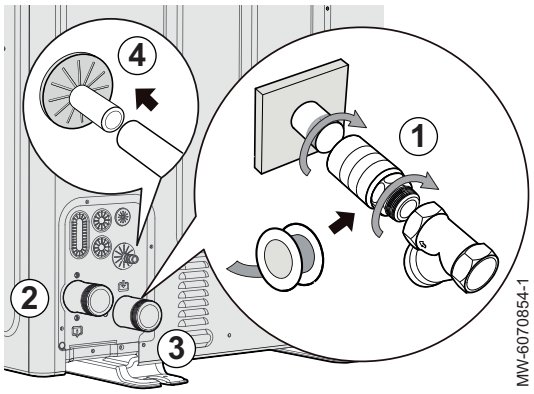
- Incorrect orientation of the water outlet and inlet can cause the outdoor unit to malfunction.
- Do not apply excessive force when connecting the pipes provided on the site, and make sure the pipes are aligned properly. Water pipe deformation can cause the unit to malfunction.
- The unit is only to be used in a closed water system.

1. Connect the Y filter of the accessory bag to the inlet of the outdoor unit. Use an extension if necessary.

Important

- It is mandatory to install a sediment filter. The Y filter is a minimum requirement. Sediment may damage the plate heat exchanger, and there could a risk of refrigerant leakage without the strainer.
- Pay attention to the correct flow direction of the Y filter.

Fig.37 Assure HP55-8-1-PHMB - Assure HP55-11-1-PHMB - Assure HP55-11-3-PHMB - Assure HP55-13-1-PHMB - Assure HP55-13-3-PHMB



2. Connect the flow pipe to the heating circuit.

Important
- For plastic pipes, make sure they are fully oxygen-tight according to DIN 4726.
- The diffusion of oxygen into the piping can lead to excessive corrosion.

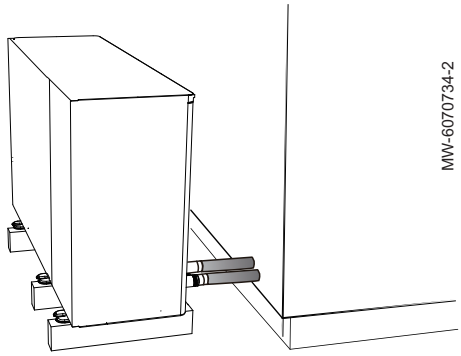
3. Connect the heating circuit return pipe.
4. Connect a hose from the safety pressure relief valve outlet to the condensate drain hole.

Important
In a glycol system, the safety pressure relief valve must be evacuated in accordance with current local standards.

6.7.6 Water pipe insulation

The complete water circuit including all pipes, must be insulated to prevent condensation during cooling operation, heating and cooling capacity reduction, and freezing of the outside water pipes in winter.

Fig.38



Important
- The insulation material should be provided with a fire resistance rating of B1 or above and comply with all applicable regulations.
- The thermal conductivity of the sealing material should be below 0.039 W/mK.

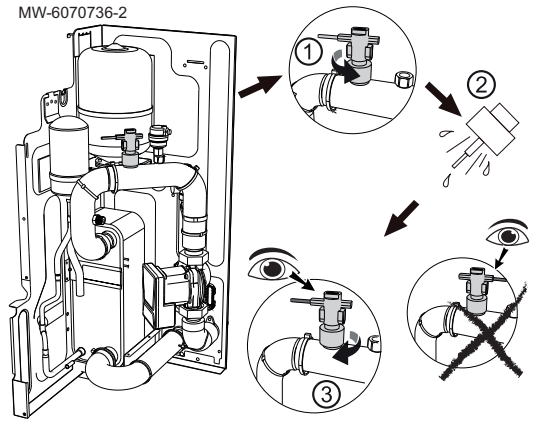
Tab.27

Length of the pipework between the outdoor unit and the indoor unit	Recommended minimum insulation thickness
< 20 m	19 mm
20~30 m	32 mm
30~40 m	40 mm
40~50 m	50 mm

If the outdoor ambient temperature is higher than 30 °C and the humidity is higher than RH 80%, the thickness of the sealing materials should be at least 20 mm to avoid condensation on the surface of the gasket.

6.7.7 Drying the flow switch

Fig.39



Water may enter the flow switch during transport and cannot be drained out, and may freeze when the temperature is low enough. The flow switch must be removed and dried before the outdoor unit is commissioned.

1. Rotate the flow switch anti-clockwise to remove it.
2. Dry the flow switch completely
3. Refit the flow switch, observing the direction of installation, with the arrow pointing towards the circulating pump.

6.8 Freeze protection

6.8.1 Software protection

The software is equipped with specific functions to protect the entire system from freezing by using the heat pump and the supplementary heater (if available).

- This control protects the water side heat exchanger from ice formation. The water-side heat exchanger electric heater is controlled according to the outdoor air temperature, the water-side heat exchanger water inlet temperature and the water-side heat exchanger water outlet temperature.
- In cooling mode, if the water inlet, water outlet or auxiliary heat source water outlet temperature is below 4 °C, the frost protection is activated. In heating/DHW mode, if the outdoor air temperature is below 3 °C and water inlet, water outlet or auxiliary heat source water outlet temperature is below 4 °C, the frost protection is activated.

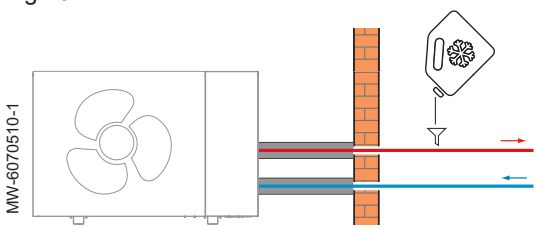
Caution
Software frost protection only works when the outdoor unit is powered.

See also
Prolonged power outage in winter, page 54

6.8.2 Frost protection

In normal operation, the outdoor unit, the indoor unit and the heating circuit are protected against frost.

Fig.40



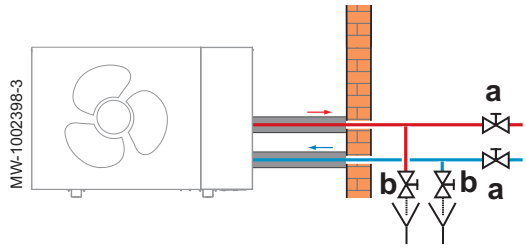
To protect the outdoor unit during prolonged power outages and outdoor temperatures below zero, one of the following solutions must be installed:

Solution using glycol
Addition of glycol to the heating circuit.

Refer to local regulations and system component manuals to check compatibility.

Manual draining solution

Fig.41



Installation of two isolation valves and two drain valves used to drain the outdoor part of the heating circuit.

a Isolation valve
b Drain valve

Caution
In the event of a prolonged power cut, manual draining is necessary.

Automatic draining solution
Installation of two anti-freeze valves on the flow and return of the heating circuit as close as possible to the outdoor unit, outside the building.

The automatic draining solution is not compatible with the use of glycol.